Menu Design for Patient–Provider Matching under Preference Uncertainty

First Author[1]

Abstract. The rise in provider turnover forces health systems to frequently rematch patients and providers, but designing matching mechanisms is challenging because patient preferences must be estimated from limited and noisy data, and matching policies are designed offline yet deployed online. We study menu design for capacity-constrained matching, where each patient is offered a set of provider options from which they select. Although we frame the problem as patient–provider matching, its structure is shared by capacity-constrained platforms that commit to offer sets before agents self-select from imperfectly estimated preferences. We prove that the general problem is weakly NP-hard, and establish an information-theoretic lower bound: any algorithm incurs regret at least $\Omega(\varepsilon \sum_j \bar{c}_j)$ in the over-subscribed regime, where ε is the noise level and $\bar{c}_j$ is provider j's capacity. Two natural baselines fail: offering a single provider per patient incurs $\Theta(\varepsilon N)$ regret, while offering all providers can perform arbitrarily poorly. To overcome these limitations, we develop the Scenario-Averaged Marginals (SAM) algorithm, a sample-average approximation that explicitly incorporates uncertainty in patient utilities and provider capacity. We prove SAM achieves $O(\varepsilon \sum_j \bar{c}_j)$ regret when the number of samples is large, matching the lower bound up to constants. On a semi-synthetic environment calibrated to a representative provider network using Medicare data, SAM improves overall utility by up to 12% over commonly used baselines while reducing geographic inequity.

Keywords: matching markets, assortment optimization, sample-average approximation, healthcare operations

1 Introduction

High provider turnover rates frequently lead patients to lose their primary care provider, which disrupts patient care and leads to worse health outcomes [29]. Today this rematching is slow and inefficient [21], handled either through ad hoc manual outreach that is administratively burdensome, or through a free-for-all in which patients select from an open directory with no coordination, leaving popular providers congested and others idle. Algorithmic matching can in principle coordinate assignments against capacity [20], but it is difficult here: preferences are not directly observed and must be estimated from limited, noisy data, and every assignment must respect scarce provider capacity.

In this work, we explore a middle path: how we can offer patients menus of providers to hedge against preference estimation uncertainty. The objective is to maximize the match quality across all patients by having them match with their preferred provider. By presenting patients with a limited set of provider options, each patient retains autonomy to select the provider that best fits their needs, even when true preferences are not fully observed by the platform (e.g., the healthcare system). The key challenge is in designing menus that optimize overall utility and balance between too much and too little choice. A menu is itself a hedge against epistemic uncertainty, and its size trades off that hedging against congestion; our goal is to design menus that balance the two.

Although we frame the problem as patient–provider matching, its structure is shared by many capacity-constrained platforms: the platform commits to offer sets before agents respond, agents self-select from what they are shown, and the platform's knowledge of preferences is imperfect and estimated from limited data. Online marketplaces, rideshare and delivery dispatch, and appointment scheduling all face the same two challenges we study: hedging against uncertain preferences, and coordinating self-interested choices against scarce capacity under an unknown response order. Common practice in these settings typically resembles the baselines we analyze, either open directories that let everyone choose freely or direct assignment that removes choice, both of which we show fail in predictable ways. We treat patient–provider matching as a concrete,

high-stakes instance of this general problem: menu design for capacity-constrained one-sided matching under epistemic preference uncertainty.

We leverage ideas from the assortment optimization literature to guide our menus [12, 30, 32]. Assortments are menus of options [5] and are commonly seen on e-commerce platforms [33]. Unlike standard assortment optimization, which models stochastic choice given known utilities, uncertainty in our setting arises from imperfect knowledge of preferences, and assortments serve as a mechanism to combat this estimation uncertainty. Moreover, in our setting, menus are designed offline, and patients respond sequentially, creating dynamic interactions through capacity constraints that are absent in standard assortment optimization [5].

We show that while the menu design problem can be optimally solved in specific instances, such as noiseless preferences and single-provider, solving the general problem remains computationally intractable. Moreover, we show that two natural and practical benchmarks perform poorly: offering at most one option per patient and offering all options to each patient. The former performs poorly because it fails to hedge against uncertainty, while the latter performs poorly because it over-allocates to high-demand options. Inspired by these failures, we develop a sample-average approximation algorithm with provable performance guarantees. Implementing this algorithm only requires the platform to estimate agent preferences, which in the healthcare setting can be obtained from a combination of patient surveys and observed patient–provider matches.

Our contributions are:

1. **Model.** We introduce a menu-based model of patient–provider matching with capacity-constrained providers, epistemic preference uncertainty, and menus committed offline before patients respond sequentially (Section 3). This offline commitment is incentive-compatible for response timing: no patient benefits from delaying, since options only weaken over time (Proposition 1).
2. **Hardness and lower bounds.** We prove that the general menu design problem is weakly NP-hard, and establish an information-theoretic lower bound showing any algorithm must incur regret $\Omega(\varepsilon \sum_j \bar{c}_j)$ (Section 4).
3. **Benchmark failures.** We show that the offer-one heuristic incurs regret $\Theta(\varepsilon N)$, and the offer-all heuristic can achieve an arbitrarily small fraction of optimal utility (Section 4).
4. **SAM algorithm with guarantees.** We develop SAM, which estimates approximate market-clearing prices on provider capacity and offers each patient their demand set at those prices. We prove SAM achieves regret $O\big((\varepsilon + \sqrt{\log(M/\nu)/S}) \sum_j \bar{c}_j\big)$ (Section 5).
5. **Empirical validation.** On a semi-synthetic environment calibrated to Medicare data, SAM improves utility by up to 12% over baselines and improves match rates for rural patients (Section 6).

2 Related Work

2.1 Patient–Provider Matching

From an algorithmic perspective, prior work on patient–provider matching typically assumes limited or no patient choice and adopts either one-shot assignment or heuristic optimization frameworks. Within the one-shot framework, prior work has developed linear programs to manage provider panel sizes [20]; deferred acceptance frameworks to perform two-stage matching [8]; and scheduling algorithms to model optimal rates of on-demand scheduling [28]. A separate line of work uses genetic programming to find fair matches [9] or balance provider workloads [36]. In contrast, we study patient–provider matching in a setting where patients retain autonomy and select providers from tailored opportunity sets. This motivates our assortment-based formulation, in which the platform controls the menu of providers offered to each patient rather than directly assigning matches.

2.2 Assortment Optimization

We adopt the assortment optimization lens, where a platform offers curated option sets and agents select from them [5]. Retailers use models to capture customer choice behavior, such as the multinomial logit [12] and nested logit [1], and generate product menus accordingly. Assortments are ubiquitous in e-commerce

settings [33] but also employed in other domains such as school choice [32], dating markets [30], restaurants [2], and rideshare [34]. Prior work spans offline (one-shot, simultaneous choice) [5] and online (sequential arrival) [4] variants, as well as settings with unknown [31], dynamically learned [7], or constrained [16] parameters. Most related is work on batch and greedy assortment [14], which find that both greedy (with small modifications) and batch policies are optimal for matching.

We differ from this literature in three ways. First, our uncertainty is *epistemic*: it arises from imperfect knowledge of patient preferences, so menus hedge against estimation error rather than modeling stochastic choice, and unlike robust assortment [31] this uncertainty is individualized and vanishes once preferences are known. Second, provider capacity couples patients' choices; unlike work on capacity under arrival uncertainty [7, 19], we know the arriving patient population and hedge only against its order, not who arrives. Third, our assortments are crafted offline while patients respond sequentially online, creating capacity-driven interactions; unlike online assortment, we cannot vary menus dynamically, reflecting healthcare settings where online variation incentivizes undesirable response patterns.

2.3 Matching Markets

By giving patients choice over their assignment, our work connects to the matching markets literature. Classical models, such as stable matching, study how preferences can be used to produce stable assignments [17], including extensions to one-sided preferences which are closer to our setting [26]. Many real-world settings feature markets with one-sided choice, such as those in school choice [10]. Our setting also connects to online matching due to the sequential arrival of patients. In bipartite online matching, optimal algorithms achieve a $1 - \frac{1}{e}$ approximation factor [24]. Generalizations include online arriving edges [15], general vertex arrivals [18], and stochastic models [13]. Unlike this literature, which optimizes over assignments, we optimize over patient menus, allowing patients to select providers subject to capacity constraints. As a result, individual choices create downstream externalities that complicate standard online matching analyses.

3 Patient–Provider Matching Model

3.1 Model Primitives

Patients and Providers. We consider a setting with N patients and M providers. Let $\theta_{i,j} \in [0,1]$ denote the utility that patient $i \in [N]$ derives from provider $j \in [M]$. Each patient also has an *exit option* (choosing no provider), denoted by an additional column $j = M+1$ with utility $\theta_{i,M+1}$. In practice, the true utility matrix $\boldsymbol{\theta}$ is not directly observed; we instead observe a noisy estimate $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + \boldsymbol{\delta}$, with $\|\boldsymbol{\delta}\|_\infty \le \varepsilon$, where $\boldsymbol{\delta}$ is zero-mean noise. Given $\hat{\boldsymbol{\theta}}$, we treat the true utilities as drawn from a distribution $\mathcal{P}_{\boldsymbol{\theta}}$ supported on $\mathcal{B} = \{\boldsymbol{\theta} \in [0,1]^{N\times(M+1)} : \|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\|_\infty \le \varepsilon\}$.

Selection and Menus. We offer a menu $\mathbf{X} \in \{0,1\}^{N\times M}$, where $X_{i,j} = 1$ indicates that provider j is offered to patient i. Here, we note that we implicitly offer the exit option $j = M+1$ to all patients. For each patient i, let L_i denote the number of offered providers on the initial menu: $L_i = \sum_{j=1}^{M} X_{i,j}$. We enforce an upper bound $L_i \le k$, where k is a hyperparameter, which mimics both logistical constraints (e.g., number of providers on the first page of a patient portal) and can mitigate option overload [11].

Response Order and Selection. Providers have limited capacities represented by the vector $\mathbf{c}_t \in \mathbb{N}^{M+1}$ at time t, where $c_{t,j}$ is the remaining capacity of provider j, and $c_{t,M+1}$ represents the unlimited exit option. Providers initially have capacity $\bar{\mathbf{c}}$, which diminishes over time. Patients in some order $\boldsymbol{\sigma}$, which is a permutation of $[N]$ drawn from a distribution $\mathcal{D}$: $\boldsymbol{\sigma} \sim \mathcal{D}$. At each step t, patient $i = \sigma_t$ selects

$$s_i = \arg\max_{j\in[M+1]} \{1[c_{t,j} > 0]\, X_{i,j}\, \theta_{i,j}\},$$

and the selected provider's capacity is updated as $c_{0,j} = \bar{c}_j, \quad c_{t+1,s_i} = c_{t,s_i} - 1$. That is, each patient selects the best available provider at each timestep. Ties are broken arbitrarily; this does not affect our results. We

assume that, conditional on the menu and remaining capacities, patient selections are deterministic (see Section 2.2 for a discussion of this). We note that just because a patient is *offered* a provider does not mean they will be *available* when it comes time to match, as patient selections deplete provider availability. We assume that patients respond truthfully; that is, they select the top option at each timestep. This is without loss of generality, as strategic patient decisions can only worsen their utility.

Optimization Problem. Our goal is to design menus $\mathbf{X}$ that maximize expected total utility across patients, accounting for stochastic response order, preference uncertainty, and capacity constraints. Formally, for a fixed menu $\mathbf{X}$, patients respond sequentially according to a random order $\boldsymbol{\sigma} \sim \mathcal{D}$, selecting their most preferred available provider from their offered menu. The platform's objective is:

$$\max_{\mathbf{X}\in\{0,1\}^{N\times M}} \; V_{\boldsymbol{\theta}}(\mathbf{X}) = \max_{\mathbf{X}} \mathbb{E}_{\boldsymbol{\sigma}\sim\mathcal{D}}\left[\sum_{i=1}^{N} \theta_{i,s_i(\mathbf{X},\boldsymbol{\sigma})}\right],$$

where $s_i(\mathbf{X}, \boldsymbol{\sigma})$ denotes the provider selected by patient i given menus $\mathbf{X}$ and response order $\boldsymbol{\sigma}$.

Solving this problem is challenging for three reasons. First, the decision space for $\mathbf{X}$ is exponential in the number of patients and providers. Second, the realized utility depends on both noisy preference estimates and random response order. Third, provider capacities induce interference across patients: assigning a provider to one patient affects the feasible options available to future patients. These features preclude direct optimization and motivate the approximation algorithms developed in Section 5.

3.2 Offline Offering and Online Response

In our setting, menus are constructed upfront and offered to all patients simultaneously, after which patients respond sequentially; the only updates thereafter are capacity-driven reductions as providers fill up. This offline-commit, online-response structure sits between two undesirable extremes. Purely offline matching assigns patients directly and eliminates choice. Purely online offerings, updated as patients arrive, restore choice but are operationally demanding: the platform must re-solve and re-issue tailored offers on every arrival, which is impractical at scale and across the long tails typical of patient response. Beyond the operational cost, dynamic menus also distort incentives. When what a patient sees depends on when they log in (e.g., providers A, B, and C on Monday and X, Y, and Z on Wednesday), delayed participation is rewarded in anticipation of better future options, further lengthening match times. Committing to menus offline removes this incentive.

Proposition 1 (Offline Menus Incentivize Early Response). *Consider the setting described above, where the menu $\mathbf{X}$ is computed offline and offered to patients sequentially according to a random order $\boldsymbol{\sigma} \sim \mathcal{D}$. For any fixed ordering σ and patient i, let $U_i(t; \sigma, \mathbf{X})$ denote the utility patient i obtains when responding at position t. Then U_i is nonincreasing in t, and hence in expectation over $\boldsymbol{\sigma} \sim \mathcal{D}$:*

$$\mathbb{E}_{\boldsymbol{\sigma}}[U_i(t; \boldsymbol{\sigma}, \mathbf{X})] \geq \mathbb{E}_{\boldsymbol{\sigma}}[U_i(t'; \boldsymbol{\sigma}, \mathbf{X})] \quad \forall\, t < t'.$$

Proof. Fix any ordering σ, and for any provider, let $g_j(\mathbf{X}, \sigma, t) = \mathbf{1}[c_{t,j}(\mathbf{X}, \sigma) > 0]$ denote whether provider j has non-zero capacity at time t, where $g_{M+1}(\mathbf{X}, \sigma, t) = 1$ for the exit option.

The utility for patient i is then: $U_i(t; \sigma, \mathbf{X}) = \max_{j\in[M+1]:g_j(\mathbf{X},\sigma,t)=1;X_{i,j}=1} \theta_{i,j}$. For fixed σ, provider availability is monotonically decreasing; $g_j(\mathbf{X}, \sigma, t) \geq g_j(\mathbf{X}, \sigma, t+1)$. Therefore, the feasible set of providers also decreases over time. Because the utility $U_i(t; \sigma, \mathbf{X})$ is a maximum over $\theta_{i,j}$, it follows that $U_i(t; \sigma, \mathbf{X}) \geq U_i(t'; \sigma, \mathbf{X}) \forall t < t'$. Taking expectations over $\boldsymbol{\sigma} \sim \mathcal{D}$ on both sides yields $\mathbb{E}_{\boldsymbol{\sigma}}[U_i(t; \boldsymbol{\sigma}, \mathbf{X})] \geq \mathbb{E}_{\boldsymbol{\sigma}}[U_i(t'; \boldsymbol{\sigma}, \mathbf{X})]$ for all $t < t'$.

4 Complexity and Benchmarks

4.1 Solution under Perfect Information

When $\varepsilon = 0$, there is no noise present, and so offering more than one option per patient is unnecessary. We can instead view the problem as a bipartite matching, and solve the following:

$$\max_{\{X_{i,j}\}} \sum_{i=1}^{N} \sum_{j=1}^{M+1} \theta_{i,j} X_{i,j} \quad \text{s.t.} \quad \sum_{j=1}^{M+1} X_{i,j} = 1 \ \forall i, \quad \sum_{i=1}^{N} X_{i,j} \le \bar{c}_j \ \forall j, \quad X_{i,j} \in \{0,1\}. \tag{1}$$

We note that there is no value in offering more than one option per patient due to the lack of uncertainty. Since menus are not needed here, we furthermore do not need to account for patient arrival order: there is no interference potential without menus.

4.2 Solution Under a Single Provider

We next characterize the optimal menu when there is a single provider ($M = 1$) with capacity $\bar{c}_1$. Because there is no competition across providers, the allocation decision becomes one-dimensional: the platform must decide which subset of patients should receive an offer to the single provider. Let $\Delta_i := \theta_{i,1} - \theta_{i,2}$ denote the utility gap between the provider (indexed by 1) and the exit option ($j = 2$). Since a patient accepts the provider if and only if $\Delta_i > 0$, the platform maximizes total utility by offering to exactly the $\bar{c}_1$ patients with the largest (positive) gaps; we return to this idea in Section 5.

Proposition 2 (Optimal Menu under Noisy Utilities for a Single Provider). *Consider the single–provider ($M = 1$) model with capacity $\bar{c}_1$. Assume that all noise terms $\delta_{i,j}$ are i.i.d. Among all deterministic menu policies that depend only on $\hat{\boldsymbol{\theta}}$, the algorithm that selects the $\bar{c}_1$ patients with the largest values of $\hat{\Delta}_i$ maximizes expected total utility.*

Proof. Fix any realization of the estimated gaps $\hat{\boldsymbol{\Delta}}$. Consider any feasible selection T of $\bar{c}_1$ patients, and suppose there exist patients $i \in T$ and $i' \notin T$ such that $\hat{\Delta}_{i'} > \hat{\Delta}_i$. Additionally, note that offering to $\bar{c}_1$ can only improve utility compared with offering to fewer than $\bar{c}_1$ patients. Note that if such a selection does not exist, then T already selects the top-$\bar{c}_1$ patients. We show that replacing i with i' cannot decrease total utility.

We note that because T consists of $\bar{c}_1$ patients, any patient $i \in T$ will receive utility $\max\{\Delta_i, 0\}$. Next, we note that $\Delta_i = (\theta_{i,1} - \theta_{i,2}) = (\hat{\theta}_{i,1} - \hat{\theta}_{i,2}) - (\delta_{i,1} - \delta_{i,2})$. Let $\delta_{i,1} - \delta_{i,2} = \bar{\delta}_i$, and note that $\bar{\delta}_i$ and $\bar{\delta}_{i'}$ have the same distribution due to the i.i.d. assumption. Next, couple $\bar{\delta}_i$ and $\bar{\delta}_{i'}$ such that $\bar{\delta}_i = \bar{\delta}_{i'}$, and note that $\Delta_{i'} = \hat{\Delta}_{i'} - \bar{\delta}_{i'} > \hat{\Delta}_i - \bar{\delta}_i = \Delta_i$. Because $\Delta_{i'} > \Delta_i$ for any realization of noise, it is better to select i' instead of i, and so T should select the top-$\bar{c}_1$ patients.

4.3 Computational Intractability

While the menu design problem can be solved optimally in limited scenarios, we next demonstrate that a general solution is intractable even when stripped down to its simplest form.

Theorem 1 (Weak NP-Hardness of Menu Optimization). *The menu optimization problem defined in Section 3 is weakly NP-hard. This holds even with only two providers and unit capacity per provider.*

We prove this via reduction from the Partition problem (full proof in Appendix C.1). Given weights $w_1, \ldots, w_N$, we construct a menu design instance with $M = 2$, $k = 1$, unit capacities, and Bernoulli noise $\theta_{i,j} \sim \text{Bernoulli}(q_i)$ where $q_i = 1 - e^{-w_i}$, with unbiased estimate $\hat{\theta}_{i,j} = q_i$. For any partition (S_1, S_2), the expected total utility equals $(1-\gamma)[2 - e^{-W_1} - e^{-W_2}] + N\gamma$ where $W_j = \sum_{i \in S_j} w_i$, which is maximized exactly when the partition is balanced, reducing menu optimization to Partition.

We next establish an information-theoretic lower bound. Let $\text{OPT}(\boldsymbol{\theta}) = \max_{\mathbf{X}} V_\theta(\mathbf{X})$ denote the welfare of an omniscient benchmark with full knowledge of the true $\boldsymbol{\theta}$. The following result shows that, against this benchmark, every algorithm must incur expected regret at least $\varepsilon \sum_{j=1}^{M} \bar{c}_j$. Since OPT is the same for all algorithms, it provides a common standard for comparing performance.

Theorem 2 (Information-Theoretic Lower Bound). *Let $\mathcal{D}$ be the uniform permutation on $[N]$. Fix any $\varepsilon \le \frac{1}{2}$ and capacities $\{\bar{c}_j\}_{j=1}^M$, and suppose the market is over-subscribed, $N \ge \alpha \sum_j \bar{c}_j$ for some constant $\alpha > 2$. Then there exists $\hat{\theta}$ and a distribution P_θ over θ, supported on $B = \{\theta : \|\theta - \hat{\theta}\|_\infty \le \varepsilon\}$, such that for any deterministic algorithm $\mathcal{A} : \hat{\theta} \mapsto \mathbf{X}$,*

$$\mathbb{E}_{\theta \sim P_\theta}\Big[\mathrm{OPT}(\boldsymbol{\theta}) - V_\theta\Big(\mathcal{A}(\hat{\boldsymbol{\theta}})\Big)\Big] \;\ge\; (1 - o(1))\varepsilon \sum_{j=1}^{M} \bar{c}_j.$$

where the $o(1)$ term vanishes as $N/\sum_j \bar{c}_j \to \infty$.

We prove this in Appendix C.1 by constructing an instance where $\alpha \bar{c}_j$ patients are competing for $\bar{c}_j$ slots for each provider. Here, $\hat{\theta}_{i,j} = \frac{1}{2}$, while $\delta_{i,j}$ is a Bernoulli random variable taking $-\varepsilon$ and ε with equal probability. The omniscient algorithm achieves a reward of $\frac{1}{2}\sum_j \bar{c}_j + \sum_{j=1}^M \sum_{i \in \text{top-}\bar{c}_j(G_j)} \delta_{i,j}$, which goes to $\frac{1}{2}\sum_j \bar{c}_j + \varepsilon \sum_j \bar{c}_j$ for large α. Meanwhile any non-omniscient algorithm receives at most $\frac{1}{2}\sum_j \bar{c}_j$, leading to the difference of $\varepsilon \sum_j \bar{c}_j$.

4.4 Two Natural Baselines: Offer-One and Offer-All

Due to the computational intractability of optimal menu design, we analyze the performance of two natural baselines; the first offers at most one provider per patient, and the second offers all providers per patient. The former corresponds to a natural offline matching approach, while the latter corresponds to common practice in healthcare [22]. We discuss when each algorithm fails, which motivates the need for better algorithms.

Offer-One A natural solution is to reduce our problem to offline matching by offering each patient at most one provider, i.e., enforcing a menu size constraint $L_i \le 1$ for all patients i. Formally, this corresponds to adopting the noiseless strategy from Section 4.1, which is optimal if $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$. While such an approach performs well when ε is small, large values of ε cause divergence between $\boldsymbol{\theta}$ and $\hat{\boldsymbol{\theta}}$. When ε is large, patient preferences in $\hat{\boldsymbol{\theta}}$ have little correlation with those in $\boldsymbol{\theta}$, making it illogical to offer only one provider based on $\hat{\boldsymbol{\theta}}$.

Proposition 3 (Offer-One Performance Characterization). *Let $\mathbf{X}^{\text{one}}$ be the menu computed from the offer-one algorithm. Then for any ε, there exists a $\boldsymbol{\theta}$ and a distribution of $\hat{\boldsymbol{\theta}}$ so $\mathbb{E}[\mathrm{OPT} - V_\theta(\mathbf{X}^{\text{one}})] = \varepsilon N$.*

Moreover, this bound is tight; for any problem instance we have that $\mathbb{E}[\mathrm{OPT} - V_\theta(\mathbf{X}^{\text{one}})] \le \varepsilon N$

We prove this by constructing an instance where noise systematically inverts the ordering between optimal and suboptimal pairs (proof in Appendix C.2). The critical issue with offer-one is its brittleness under preference uncertainty, which motivates algorithms that offer multiple providers per patient to hedge against uncertainty.

Offer-All Another natural solution is to offer every patient the largest menu allowed: all providers when $k = M$ (i.e., $\mathbf{X} = \mathbf{1}$), or their top-k by estimated utility when $k < M$. This is equivalent to having a central available provider directory and allowing all patients to access and select freely. While such a policy is appealing, it performs poorly because providers can be over-demanded, leading to inter-patient competition.

Proposition 4 (Offer-All Performance Characterization). *Let $\mathbf{X}^*$ be the optimal menu and let $\mathbf{X}^{\text{all}} = \mathbf{1}$ be the offer-all menu. For any $\alpha \in (0,1)$, there exists $\boldsymbol{\theta}$ so that $V_\theta(\mathbf{X}^{\text{all}}) \le \alpha V_\theta(\mathbf{X}^*)$.*

Proof. Let $\varepsilon = 0$ and $N = M$. Fix a single pair (i^*, j^*) and set $\theta_{i^*,j^*} = 1$, $\theta_{i,j^*} = 2\gamma$ for all $i \ne i^*$, and $\theta_{i,j} = \gamma$ for all other i, j pairs, with $\gamma \ll 1$. Every patient therefore strictly prefers provider j^*, so under offer-all all N patients contend for it; since $\bar{c}_{j^*} = 1$, provider j^* goes to the first arrival. The optimal policy achieves a total utility of $1 + \gamma(N-1)$, while the offer-all policy achieves a utility of at most $\frac{1}{N} + \gamma(N+1)$ because i^* and j^* are matched with probability $\frac{1}{N}$ (which is the probability patient i^* arrives first). For a given α, solving for γ in $\alpha = \frac{1/N + \gamma(N+1)}{1+\gamma(N-1)}$ yields the corresponding $\boldsymbol{\theta}$. As $\gamma \to 0$ the ratio tends to $1/N$, so $\alpha = O(1/N)$, and for any target $\alpha \in (0,1)$ it suffices to take $N = M$ large enough and tune γ. The guarantee thus degrades without bound as the market grows, since a single unit-capacity provider must serve a growing population.

Any algorithm needs to address two critical faults of these two baselines: 1) offering a single provider per patient is brittle under preference uncertainty, and 2) offering all providers ignores downstream capacity congestion and inter-patient competition. We address both through a new algorithm in Section 5.

5 Scenario-Averaged Marginals (SAM) Algorithm

In this section, we introduce the Scenario-Averaged Marginals (SAM) algorithm to tackle this problem. We develop a sample-average approximation (SAA) approach that samples utility realizations to address preference uncertainty, then uses LP dual variables from each scenario to estimate the opportunity cost of scarce providers. The resulting algorithm constructs menus via a tractable scoring rule and admits performance guarantees under mild regularity conditions. We prove performance guarantees for SAM that depend on a structural $O(\varepsilon)$ term and a sample-complexity term that can be driven to zero with many samples.

5.1 Stochastic Optimization for Menu Design

Menu design is challenging because neither patient utility $\boldsymbol{\theta}$ nor ordering $\boldsymbol{\sigma}$ are known a priori. To overcome this, we study an approximation of the problem built on top of sample estimates of $\boldsymbol{\theta}$ and $\boldsymbol{\sigma}$, then optimize the average performance across these samples. Because the total utility is bounded, when the number of samples is large enough, performance on the sample-average problem should converge to the true problem. Formally, we consider S paired samples $\{\boldsymbol{\theta}^{(s)}, \boldsymbol{\sigma}^{(s)}\}_{s=1}^{S}$, where each pair is sampled i.i.d. from $\mathcal{P}_{\boldsymbol{\theta}} \times \mathcal{D}$. Using these samples, we transform the original problem, $R(\mathbf{X})$ into the SAA problem $\hat{R}_S(\mathbf{X})$:

$$R(\mathbf{X}) := \mathbb{E}_{\boldsymbol{\theta}\sim\mathcal{P}_{\boldsymbol{\theta}},\boldsymbol{\sigma}\sim\mathcal{D}}[V_{\boldsymbol{\theta}}(\mathbf{X};\sigma)], \quad \hat{R}_S(\mathbf{X}) := \frac{1}{S}\sum_{s=1}^{S} V_{\boldsymbol{\theta}^{(s)}}(\mathbf{X}, \boldsymbol{\sigma}^{(s)}).$$

Let $y_{i,j,s}$ represent a binary variable indicating if patient i selects provider j in scenario s. We can find the optimal assortment $\mathbf{X}$ by solving the following SAA mixed-integer linear program (SAA-MILP):

$$\begin{aligned}
\max_{\{X_{i,j},y_{i,j,s}\}} \quad & \sum_{s=1}^{S}\sum_{i=1}^{N}\sum_{j=1}^{M+1} y_{i,j,s}\theta_{i,j}^{(s)} \\
\text{s.t.} \quad & \textstyle\sum_{j=1}^{M} X_{i,j} \le k, \;\; \sum_{j=1}^{M+1} y_{i,j,s} = 1, \;\; y_{i,j,s} \le X_{i,j} \quad \forall i \in [N], j \in [M], s \in [S] \\
& \textstyle\sum_{i=1}^{N} y_{i,j,s} \le \bar{c}_j, \quad c_{t,j,s} = \bar{c}_j - \sum_{t'<t} y_{\sigma_{t'}^{(s)},j,s} \quad \forall t \in [N], j \in [M], s \in [S] \\
& \textstyle\sum_{j'=1}^{M} y_{\sigma_t^{(s)},j',s}\theta_{\sigma_t^{(s)},j',s} \ge \theta_{\sigma_t^{(s)},j,s} \mathbb{1}[c_{t,j,s} > 0]\, X_{\sigma_t^{(s)},j} \quad \forall t \in [N], j \in [M], s \in [S] \\
& X_{i,j}, y_{i,j,s} \in \{0,1\}, \quad c_{t,j,s} \ge 0
\end{aligned} \tag{2}$$

5.2 Dual Variables for Scarce Providers

Equation 2 is computationally intractable. Because all S scenarios are coupled, solving the MILP requires solving $O(NMS)$ variables, and even its continuous relaxation remains impractical to solve at real-world problem scales. We therefore seek a tractable approximation.

There are two sources of uncertainty in our setting: patient preferences $\boldsymbol{\theta}$ and $\boldsymbol{\sigma}$. These contribute asymmetrically to overall performance. Arrival order affects which patients encounter depleted capacity. Preference uncertainty, by contrast, directly determines which providers are offered to which patients and can lead to systematically poor menus when estimates are noisy. While in the worst case, adversarial arrival orders can modify performance by up to $\Theta(N)$, in practice, we find that performance concentrates (see Figure 1); we suspect this occurs because preferences are heterogeneous, mitigating the impact of adversarial orderings. Moreover, the menu $\mathbf{X}$ can be constructed to hedge against preference uncertainty (by offering more options to patients), while no equivalent exists for arrival-order uncertainty. As a result, we focus on algorithms to tackle variations in $\boldsymbol{\theta}$, while treating $\boldsymbol{\sigma}$ as fixed.

Algorithm 1 Scenario-Averaged Marginals (SAM) Algorithm

Input: Estimated utilities $\hat{\theta}$, capacities $\bar{c}$, number of scenarios S, maximum menu size k
Output: Menu decision matrix $X \in \{0,1\}^{N \times M}$
for $s = 1$ to S **do**
  Sample perturbed utilities $\boldsymbol{\theta}^{(s)}$ and let $\Delta_{i,j}^{(s)} \leftarrow \theta_{i,j}^{(s)} - \theta_{i,M+1}^{(s)}$ for all i, j
  Let $\hat{\lambda}_{j,s}$ be the dual variable for constraint $\sum_{i=1}^{N} y_{i,j} \leq \bar{c}_j$
end for
$\hat{\lambda}_j^{\mathrm{SAM}} \leftarrow \frac{1}{S} \sum_{s=1}^{S} \hat{\lambda}_{j,s}$ for all j
$m_{i,j} \leftarrow \left(\frac{1}{S} \sum_{s=1}^{S} \max\{\Delta_{i,j}^{(s)} - \hat{\lambda}_j^{\mathrm{SAM}}, 0\} \right)$
for each patient i **do**
  Let A_i be the k providers with largest positive $m_{i,j}$
  for each provider $j \in [M]$ **do**
    $X_{i,j} \leftarrow \mathbb{1}\{j \in A_i\}$
  end for
end for
return X

We first focus on the case with only $\hat{\boldsymbol{\theta}}$ and show that a threshold-based policy is optimal for a simplified version of the problem, where we select $y_{i,j}$ directly to maximize $\sum_{i,j} y_{i,j}\theta_{i,j}$. Here, the optimal policy has the form $y_{i,j} = 1$ if $\Delta_{i,j} - \lambda_j > 0$, where λ_j are the dual variables corresponding to capacity.

Lemma 1 (Optimality of Dual Variables). *Fix any utility matrix $\theta \in [0,1]^{N \times (M+1)}$ and capacities $\bar{c}_j$, and let $\Delta_{i,j} := \theta_{i,j} - \theta_{i,M+1}$. Consider*

$$\max_{y \geq 0} \sum_{i,j} y_{i,j}\theta_{i,j} \quad s.t. \quad \sum_{j} y_{i,j} = 1 \ \forall i, \quad \sum_{i} y_{i,j} \leq \bar{c}_j \ \forall j \in [M].$$

Let λ_j be optimal dual prices for the capacity constraints. Then an optimal solution assigns patient i to

$$j^*(i) \in \arg\max_{j \in [M]} (\Delta_{i,j} - \lambda_j) \quad \text{if } \max_j (\Delta_{i,j} - \lambda_j) > 0, \qquad \text{otherwise exits},$$

i.e., $y_{i,j}^* = 1$ for $j = j^*(i)$ in the first case, $y_{i,M+1}^* = 1$ in the second.

Proof. Existence of $\boldsymbol{\lambda}$ follows from LP duality. By dual feasibility / stationarity, for every (i, j), $j \in [M+1]$ (with $\lambda_{M+1} = 0$ due to abundant capacity for the exit option):

$$\theta_{i,j} - \omega_i - \lambda_j \leq 0, \quad \text{with equality iff } y_{i,j}^* > 0,$$

where ω_i is the dual of $\sum_j y_{i,j} = 1$. Rearranging, $\omega_i \geq \theta_{i,j} - \lambda_j$ for every j, with equality exactly at the assigned option, so $\omega_i = \max_{j \in [M+1]}(\theta_{i,j} - \lambda_j)$. Subtracting $\theta_{i,M+1}$ from the maximization itself, i.e. comparing every option to exit rather than to zero, and writing $\Delta_{i,j} := \theta_{i,j} - \theta_{i,M+1}$ (so exit's own score becomes $\Delta_{i,M+1} = 0$):

$$\omega_i - \theta_{i,M+1} = \max_{j \in [M+1]} \left(\Delta_{i,j} - \lambda_j\right) = \max\left(0, \ \max_{j \in [M]} (\Delta_{i,j} - \lambda_j)\right).$$

Suppose $y_{i,j_0}^* = 1$ for some j_0. By complementary slackness, $\omega_i - \theta_{i,M+1} = \Delta_{i,j_0} - \lambda_{j_0}$. For every other j, dual feasibility only gives $\Delta_{i,j} - \lambda_j \leq \omega_i - \theta_{i,M+1}$. So j_0 must achieve the maximum of $\Delta_{i,j} - \lambda_j$ over $j \in [M+1]$. If $\max_{j \in [M]}(\Delta_{i,j} - \lambda_j) > 0$, some provider beats exit's score of 0, so the assigned option is some $j^*(i) \in \arg\max_{j \in [M]}(\Delta_{i,j} - \lambda_j)$. Otherwise, exit's score of 0 is the maximum and $y_{i,M+1}^* = 1$.

5.3 Scenario-Averaged Marginals (SAM) Algorithm

We introduce **Scenario-Averaged Marginals (SAM)**, an algorithm that estimates $\boldsymbol{\lambda}$ and uses this to compute assortments (Algorithm 1). $\boldsymbol{\lambda}$ captures the opportunity cost of matching with a particular provider, and informally represents the value of increasing $\bar{c}_j$ by one. If we can accurately estimate $\boldsymbol{\lambda}$, then we can set $X_{i,j} = \mathbb{1}[\Delta_{i,j} - \lambda_j > 0]$ so patients are only offered providers with positive surplus. The key challenge is accurately estimating $\boldsymbol{\lambda}$ from S samples of $\boldsymbol{\theta}^{(s)}$. Due to the intractability of Equation 2, we instead solve S independent problems, resulting in estimates $\hat{\boldsymbol{\lambda}}_{\cdot,s}$, which we then aggregate as $\hat{\lambda}_j^{\text{SAM}} = \frac{1}{S}\sum_{s=1}^{S} \hat{\lambda}_{j,s}$. Once we can estimate $\hat{\boldsymbol{\lambda}}$, we let $m_{i,j} = \frac{1}{S}\sum_{s=1}^{S} \max\{\Delta_{i,j}^{(s)} - \hat{\lambda}_j^{\text{SAM}}, 0\}$, then let $X_{i,j} = 1$ for the top k largest positive $m_{i,j}$. Equivalently, λ are market-clearing prices on capacity and each patient is offered their demand set at those prices. SAM solves S independent bipartite matching problems, one per scenario, each solvable in polynomial time by standard methods. This results in runtime scaling linearly in S and near-linearly in N.

5.4 Theoretical Guarantees

To demonstrate performance guarantees, we first introduce three regularity conditions; these regularity conditions operate on the smoothed empirical distribution of utilities arising from noisy samples.

1. **R1 - Twice Differentiability** - We require that the empirical density of patient utilities, after smoothing by noise, is twice differentiable and continuous near the assignment boundary. In practice, this holds as long as the noise distribution is continuous, such as Gaussian noise added to patient utilities. Continuity and differentiability is necessary for Taylor approximation. In practice, such a condition would break if noise were discrete, which could create point masses.
2. **R2 - Non-Singularity** - Let $H_j(\lambda)$ be the empirical market share that provider j gets, smoothed over S noisy draws; that is the fraction of patients whose most preferred option, after accounting for provider capacity (λ_j) is j. We require that the Jacobian, $DH(\boldsymbol{\lambda}^*)$, is non-singular. This is required so that we can solve and bound $\|\boldsymbol{\lambda}^* - \hat{\boldsymbol{\lambda}}^{\text{SAM}}\|_\infty$. This holds as long as providers are not perfect substitutes for one another, so that no two rows of $DH(\boldsymbol{\lambda}^*)$ are identical. This is unlikely to fail unless providers attract nearly identical patient populations.
3. **R3 - Bounded Derivatives** - We require that the partial sums for second and third derivatives of H_j are bounded. Bounded derivatives arise when patient utilities vary smoothly with provider characteristics, so that small changes in prices do not induce discontinuous changes in demand.

R1–R3 are standard smoothness and non-degeneracy conditions in the discrete-choice and market-equilibrium literature (analogous to regularity conditions used to establish asymptotic normality of estimated equilibrium prices); they are satisfied whenever the noise distribution has a bounded, smooth density, which holds for common choices such as Gaussian or Laplace noise. We verify these regularity conditions empirically in Appendix B.3. We next show that under the regularity conditions, $\hat{\boldsymbol{\lambda}}^{\text{SAM}}$ and $\boldsymbol{\lambda}^*$ differ due to three quantities: the first is an $O(\varepsilon^2)$ error term due to preference uncertainty, and the second is a $O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right)$ which arises because we estimate from finite samples, and the third $O(1/\sqrt{N})$ term that is subsumed for $N \geq 1/\varepsilon^2$.

Lemma 2 (Dual Estimation Bias is Second-Order). *Let*

$$H_j(\lambda) := \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\Delta_{i,j} - \lambda_j = \max\left(0,\ \max_{j'}(\Delta_{i,j'} - \lambda_{j'})\right)\right]$$

be the empirical market share of provider j at prices λ, evaluated on the true gaps Δ. Let λ^* satisfy $H_j(\lambda^*) = \rho_j$ for all $j \in J^*$, where $J^* := \{j \in [M] : \lambda_j^* > 0\}$. Suppose that regularity conditions R1-R3 hold. Let $\hat{\lambda}_j^{\text{SAM}} = \frac{1}{S}\sum_{s=1}^{S} \hat{\lambda}_{j,s}$, where $\hat{\lambda}_{j,s}$ is the dual variable for provider j in scenario s. If $\mathbb{E}[\delta_{i,j}] = \mathbb{E}[\delta_{i,j}^{(s)}] = 0$ with i.i.d. noise, then with probability at least $1 - \nu$ over the S sampled scenarios,

$$\|\hat{\lambda}^{\text{SAM}} - \lambda^*\|_\infty \leq O(\varepsilon^2) + O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right) + O(1/\sqrt{N}).$$

The $O(\varepsilon^2)$ term arises from a Taylor expansion of the market-share function showing that zero-mean noise introduces only second-order bias in the dual prices. The $O(\sqrt{\log M/S})$ term follows from a Hoeffding bound over the S sampled scenarios (full proof in Appendix C.3).

We next show that this implies a performance bound for the overall performance of SAM:

Theorem 3 (Performance Guarantee for SAM). *Consider a population of N patients with true utilities θ and noisy utilities $\hat{\theta}$. Let*

$$H_j(\lambda) := \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\Delta_{i,j} - \lambda_j = \max\left(0,\ \max_{j'}(\Delta_{i,j'} - \lambda_{j'})\right)\right]$$

denote the empirical market share of provider j at prices λ, and let λ^* satisfy $H_j(\lambda^*) = \rho_j$ for all $j \in J^*$. Suppose that regularity conditions R1-R3 hold. If $\sum_j \bar{c}_j \le N$, $N \ge 1/\varepsilon^2$ (i.e., the market is not very thin relative to ε) and i.i.d with zero mean., then with probability at least $1-\nu$ over the S sampled scenarios,

$$\mathrm{OPT} - R(\mathbf{X}^{\mathrm{SAM}}) \ \le\ O\left((\varepsilon + \sqrt{\frac{\log(M/\nu)}{S}})\sum_j \bar{c}_j\right).$$

Proof (Proof sketch). Full proof in Appendix C.3. By LP strong duality, $\mathrm{OPT} = \sum_i \theta_{i,M+1} + \sum_i \max(0, \max_j(\Delta_{i,j} - \lambda_j^*)) + \sum_j \bar{c}_j \lambda_j^*$. We compare OPT against SAM's threshold policy $\mathbf{X}^{\hat{\Delta},\hat{\lambda}}$ (which includes provider j in patient i's menu iff $\hat{\Delta}_{i,j} \ge \hat{\lambda}_j$), then bound the additional loss from the top-k cap.

Near-threshold pairs. Call a pair (i,j) *safe-in* if $\Delta_{i,j} > \lambda_j^* + O(\varepsilon)$ and *safe-out* if $\Delta_{i,j} < \lambda_j^* - O(\varepsilon)$. By Lemma 2, $\|\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\|_\infty \le O(\varepsilon^2)$ (ignoring the sample-complexity term, with $N \ge 1/\varepsilon^2$), so safe-in pairs are always included and safe-out pairs are always excluded. Any discrepancy between OPT and SAM is confined to *near-threshold* pairs with $|\Delta_{i,j} - \lambda_j^*| = O(\varepsilon)$, each contributing at most $O(\varepsilon)$ welfare loss.

Counting near-threshold pairs. By the capacity constraint, at most $\bar{c}_j$ patients can be mis-classified at provider j. Summing over j gives total threshold error $O(\varepsilon \sum_j \bar{c}_j)$.

Top-k cap. The top-k truncation only displaces the optimal provider $j^*(i)$ if a competing provider lies within $O(\varepsilon)$ of $j^*(i)$ in true score, contributing at most $O(\varepsilon)$ per affected patient. Since at most $\sum_j \bar{c}_j$ patients have a positive argmax, this adds $O(\varepsilon \sum_j \bar{c}_j)$ total.

Finite-S correction. For finite S, Lemma 2 gives the general dual error , which widens the near-threshold band and contributes $O(\sum_j \bar{c}_j \cdot \sqrt{\log(M/\nu)/S}) + O(\sum_j \bar{c}_j/\sqrt{N})$ to the welfare gap. Under the hypothesis $N \ge 1/\varepsilon^2$ of Theorem 3, the last term satisfies $O(\sum_j \bar{c}_j/\sqrt{N}) \le O(\varepsilon \sum_j \bar{c}_j)$, so it is dominated by the structural $O(\varepsilon \sum_j \bar{c}_j)$ term already counted above and can be dropped.

Theorem 2 shows that regret approaching $\varepsilon \sum_j \bar{c}_j$ is unavoidable in the over-subscribed limit, while Theorem 3 shows that SAM attains $O(\varepsilon \sum_j \bar{c}_j)$ under the same i.i.d. zero-mean noise; once $S \gtrsim \log(M/\nu)/\varepsilon^2$, the sampling term is dominated and SAM matches the lower bound up to constants. Against offer-one, whose worst-case error is εN (Proposition 3), SAM's regret is of the same order and strictly better when capacity is scarce ($\sum_j \bar{c}_j \ll N$): the relative advantage is the scarcity factor $\sum_j \bar{c}_j/N$, and the absolute gap $\varepsilon(N - \sum_j \bar{c}_j)$ grows with the noise level ε. Against offer-all, which achieves at most $\alpha\, V_\theta(\mathbf{X}^*)$ for any $\alpha \in (0,1)$ (Proposition 4), SAM can be arbitrarily better, since offer-all ignores provider capacity. Finally, SAM realizes its worst-case regret only when many patient–provider pairs are near-indifferent at the clearing prices ($\Delta_{i,j} \approx \lambda_j^*$) at once, a configuration that is unlikely under realistic preference distributions.

6 Empirical Results

Experimental Setup We construct a semi-synthetic simulation to better understand how operational factors impact performance in practice. We select a default set of hyperparameters to mimic the patient and provider counts in a real-world health system, and let $N = 1225$ and $M = 700$. We select default values for other

parameters as: $\mathcal{D}$ is uniform over permutations of $[N]$, $k = 25$, $S = 10$, $\varepsilon = 0.25$, $\bar{c}_j \sim \text{Poisson}(\lambda)$, where we let $\lambda = 1$ by default. We construct $\boldsymbol{\theta}$ as a proxy for real patient preferences by incorporating two key factors: geographic proximity and comorbidity alignment. Prior work demonstrates that geographic proximity is the top factor impacting patient utilities [6, 35], with patients preferring closer providers to further away ones. We describe the exact parameterization of both parameters in Appendix A.

Throughout our experiments, we consider three baselines: offer-one, offer-all, and random ($\mathbf{X}_i$ is a random uniform k subset of providers). In each experiment, we construct menus according to these algorithms and then simulate sequential patient selection of providers according to their true θ. For each run, we sample a new random permutation σ, and compute metrics for patient utilities. All experiments are averaged across nine seeds, with 25 permutations of σ per seed. For all figures, we report the standard error.

Main Comparison We first compare algorithms along three metrics. First, we consider the average utility, normalized by the performance by the omniscient algorithm. We quantify the quality of choices offered by an algorithm through the choice utility. For a patient i presented with r options, we define the choice utility as the ratio of the total utility of the offered options to the sum of the top-r values of $\theta_{i,j}$: that is, the best possible utility achievable with r options. Finally, to capture fairness, we compute the match rate for the area (ZIP code) with the 25th percentile population size; this captures how well the algorithm matches patients in rural areas, and whether it is fair across geographic areas.

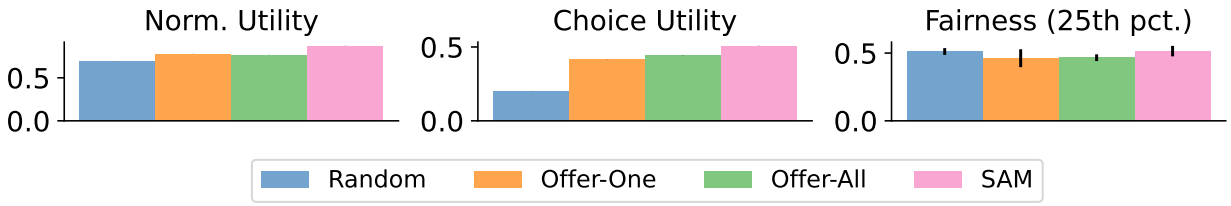


Fig. 1: Performance of the algorithms across the overall utility, level of choice offered, and fairness. SAM improves the average utility by 12% and choice utility by 14% relative to the best baseline, while maintaining comparable fairness with other baselines (note that standard error is less than 0.1% here).

Figure 1 demonstrates that SAM improves utility, choice (in terms of the choice utility) and maintains geographical equity (in terms of the utility for the 25th percentile zipcode, in terms of population). In particular, we find that SAM improves the average utility by 12% and choice utility by 14% compared to the best baseline. A 12% increase in average utility corresponds to patients being matched with providers meaningfully closer to them or more concordant with their comorbidities, which can have a positive impact on medication adherence and care continuity [3]. Additionally, SAM is within 13% of the omniscient optimal policy, and so achieves at least a 0.87-approximation factor. In Appendix B.4 we show that similar results hold under an MNL choice model, in Appendix B.2 we show that similar results hold across patient–provider ratios, and in Appendix B.1, we show that $S = 10$ samples suffices for good performance.

Effect of Noise Figure 2 compares the algorithms when varying the noise level ε. We first compare against the optimal MILP solution, which is feasible at small scale. SAM achieves an approximation ratio above 0.95, while offer-one and offer-all degrade with noise. For the standard $N = 1225$ scenario, we compute the normalized utility, and find that SAM has the biggest improvement over baselines when ε is large, as it is 13% better than alternatives at $\varepsilon = 0.5$. When ε is small, the offer-one algorithm can successfully coordinate patients and providers because the utility is perfectly known. As noise grows, SAM's menu-based hedging provides a meaningful advantage, which is meaningful because utility estimates are imperfect in practice.

Geographic Equity Beyond menu quality, we also examine the geographic fairness of the SAM algorithm by measuring the distribution of match rates according to ZIP codes. In patient–provider matching, ensuring

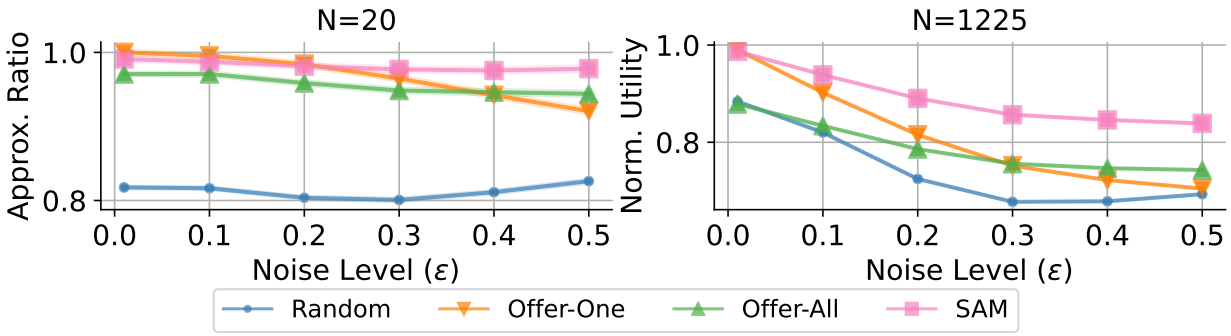


Fig. 2: (left) Approximation ratio for algorithms compared to the optimal SAA-MILP algorithm when varying the noise level ε and (right) normalized utilities when varying noise level. SAM maintains the largest advantage compared to baselines for large noise, while offer-one and offer-all perform worse with increasing noise.

Method	25th Pct. Match Rate	Gini Coefficient
Random	0.46 ± 0.02	0.15 ± 0.00
Offer-One	0.34 ± 0.06	0.21 ± 0.01
Offer-All	0.45 ± 0.03	0.11 ± 0.00
SAM	0.47 ± 0.04	0.14 ± 0.01

Table 1: Fairness metrics across algorithms: 25th percentile match rate (higher is better) and Gini coefficient (lower is better) for ZIP-code level outcomes.

geographic equity is important so rural patients also get serviced at high rates. We think of population as a dimension that impacts match quality, and similar analyses can be repeated for other fairness-related quantities. We focus on ZIP code population density as a proxy for rurality, since lower-density areas typically have fewer nearby providers and face greater barriers to matching.

Quantitatively, we can capture fairness through two metrics: the 25th percentile outcome and the Gini coefficient (Table 1). The 25th percentile outcome captures whether rural ZIP codes maintain high match rates. The Gini coefficient is a measure of spread among the match rates for different ZIP codes, with lower Gini coefficient indicating a fairer distribution. We find that the offer-all achieves the lowest Gini coefficient, with SAM close behind and random and offer-one worse, and that the SAM algorithm achieves the higher 25th percentile fairness. While the offer-all algorithm achieves a lower Gini coefficient than SAM, it does so by reducing match rates for urban areas, which makes outcomes uniformly worse. Conversely, SAM redistributes matches so those in rural ZIP codes benefit without harming urban areas, which leads to a higher match rate for the 25th percentile ZIP code. In particular, SAM increases match rates for rural ZIP codes, with a 38% gain over offer-one and a 4% gain over offer-all for the 25th percentile ZIP code by population.

7 Conclusion

A carefully designed menu of providers hedges against preference uncertainty while managing capacity constraints, improving utility by up to 12% over standard baselines. The key insight is that structured, capacity-aware choice is itself a hedge against noise: tailoring menus produces good matches even when preferences are poorly estimated, with provable performance guarantees. We also find improved geographic equity as a byproduct, since high-quality providers are spread across patients rather than concentrated. Finally, this structure extends beyond healthcare: capacity-constrained platforms from online marketplaces to rideshare dispatch to appointment scheduling commit to offer sets before agents self-select and must estimate preferences, and SAM offers a concrete template for hedging uncertainty and coordinating choice.

Bibliography

[1] Alfandari, L., Hassanzadeh, A., Ljubić, I.: An exact method for assortment optimization under the nested logit model. European Journal of Operational Research **291**(3), 830–845 (2021)

[2] Alptekinoglu, A., Dhanorkar, S., Zou, F.: Effect of product variety on food waste: A consumer-level analysis. Available at SSRN 5363717 (2025)

[3] Alsan, M., Garrick, O., Graziani, G.: Does diversity matter for health? experimental evidence from oakland. American Economic Review **109**(12), 4071–4111 (2019)

[4] Aouad, A., Saban, D.: Online assortment optimization for two-sided matching platforms. Management Science **69**(4), 2069–2087 (2023)

[5] Bertsimas, D., Mišic, V.V.: Data-driven assortment optimization. Management Science **1**, 1–35 (2015)

[6] Buzza, C., Ono, S.S., Turvey, C., Wittrock, S., Noble, M., Reddy, G., Kaboli, P.J., Reisinger, H.S.: Distance is relative: unpacking a principal barrier in rural healthcare. Journal of general internal medicine **26**, 648–654 (2011)

[7] Caro, F., Martínez-de Albéniz, V., Rusmevichientong, P.: The assortment packing problem: Multiperiod assortment planning for short-lived products. Management Science **60**(11), 2701–2721 (2014)

[8] Chen, R., Chen, M., Yang, H.: Dynamic physician-patient matching in the healthcare system. In: 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC). pp. 5868–5871. IEEE (2020)

[9] Chen, X., Zhao, L., Liang, H., Lai, K.K.: Matching patients and healthcare service providers: a novel two-stage method based on knowledge rules and owa-nsga-ii algorithm. J. Comb. Optim. **37**(1), 221–247 (Jan 2019). https://doi.org/10.1007/s10878-017-0221-2, https://doi.org/10.1007/s10878-017-0221-2

[10] Chen, Y., Sönmez, T.: School choice: an experimental study. Journal of Economic theory **127**(1), 202–231 (2006)

[11] Chernev, A., Böckenholt, U., Goodman, J.: Choice overload: A conceptual review and meta-analysis. Journal of Consumer Psychology **25**(2), 333–358 (2015)

[12] Davis, J., Gallego, G., Topaloglu, H.: Assortment planning under the multinomial logit model with totally unimodular constraint structures. Work in Progress (2013)

[13] Dynkin, E.B.: The optimum choice of the instant for stopping a markov process. Soviet Mathematics **4**, 627–629 (1963)

[14] Eom, M., Toriello, A.: Batching and greedy policies: How good are they in dynamic matching? Manufacturing & Service Operations Management (2025)

[15] Epstein, L., Levin, A., Segev, D., Weimann, O.: Improved bounds for online preemptive matching. arXiv preprint arXiv:1207.1788 (2012)

[16] Feldman, J., Topaloglu, H.: Capacitated assortment optimization under the multinomial logit model with nested consideration sets. Operations Research **66**(2), 380–391 (2018)

[17] Gale, D., Shapley, L.S.: College admissions and the stability of marriage. The American mathematical monthly **69**(1), 9–15 (1962)

[18] Gamlath, B., Kapralov, M., Maggiori, A., Svensson, O., Wajc, D.: Online matching with general arrivals. In: 2019 IEEE 60th Annual Symposium on Foundations of Computer Science (FOCS). pp. 26–37. IEEE (2019)

[19] Golrezaei, N., Nazerzadeh, H., Rusmevichientong, P.: Real-time optimization of personalized assortments. Management Science **60**(6), 1532–1551 (2014)

[20] Harrington, W., Rubin, P.A., Bai, L.: An optimization approach to panel size management. Operations Research for Health Care **31**, 100313 (2021)

[21] Hedden, L., Ahuja, M.A., Lavergne, M.R., McGrail, K.M., Law, M.R., Cheng, L., Barer, M.L.: How long does it take patients to find a new primary care physician when theirs retires: a population-based, longitudinal study. Human Resources for Health **19**, 1–11 (2021)

[22] Herwartz, H., Strumann, C.: Too many cooks could spoil the broth: choice overload and the provision of ambulatory health care. International Journal of Health Economics and Management **24**(3), 357 (2024)
[23] Johnson, D.S., Garey, M.R.: Computers and intractability: A guide to the theory of NP-completeness. WH Freeman (1979)
[24] Karp, R.M., Vazirani, U.V., Vazirani, V.V.: An optimal algorithm for on-line bipartite matching. In: Proceedings of the twenty-second annual ACM symposium on Theory of computing. pp. 352–358 (1990)
[25] Medicare & Medicaid: Doctors and clinicians data. `https://data.cms.gov/provider-data/topics/doctors-clinicians` (2025), accessed: 2025-02-02
[26] Park, S.: Stable marriage with one-sided preference. arXiv preprint arXiv:2401.03231 (2024)
[27] Piccirillo, J.F., Vlahiotis, A., Barrett, L.B., Flood, K.L., Spitznagel, E.L., Steyerberg, E.W.: The changing prevalence of comorbidity across the age spectrum. Critical reviews in oncology/hematology **67**(2), 124–132 (2008)
[28] Qu, X., Rardin, R.L., Williams, J.A.S., Willis, D.R.: Matching daily healthcare provider capacity to demand in advanced access scheduling systems. European Journal of Operational Research **183**(2), 812–826 (2007)
[29] Reddy, A., Pollack, C.E., Asch, D.A., Canamucio, A., Werner, R.M.: The effect of primary care provider turnover on patient experience of care and ambulatory quality of care. JAMA internal medicine **175**(7), 1157–1162 (2015)
[30] Rios, I., Torrico, A.: Platform design in matching markets: a two-sided assortment optimization approach. arXiv preprint arXiv:2308.02584 (2023)
[31] Rusmevichientong, P., Topaloglu, H.: Robust assortment optimization in revenue management under the multinomial logit choice model. Operations research **60**(4), 865–882 (2012)
[32] Shi, P.: Assortment planning in school choice. Tech. rep., mimeo (2016)
[33] Tian, Q., Tang, H., Wang, K.: Enhancing customer baskets through assortment optimization. Journal of Engineering **2**(3), 29 (2024)
[34] Yang, Y., Umboh, S.W., Ramezani, M.: Freelance drivers with a decline choice: Dispatch menus in on-demand mobility services for assortment optimization. Transportation Research Part B: Methodological **190**, 103082 (2024)
[35] Yen, W., Mounts, T.: Washington state health services research project. Washington State Office of Financial Management (2013)
[36] Zhu, W., He, J., Guo, H.: Doctor-patient bilateral matching considering diagnosis and treatment perception in the absence of public health resources. Frontiers in Public Health **10**, 1094523 (2023)

A Experimental Details

We estimate the patient and provider sizes as $N = 1225$ and $M = 700$ from the average panel size and provider count of the organization. For the response order, we let $\sigma \sim \mathcal{D}$, where $\mathcal{D}$ is the uniform distribution over permutations. We construct θ to reflect geographic proximity and the presence of comorbidities. Geographic proximity is the top factor impacting patient match quality [6, 35], while patients with comorbidities are best served by providers with the corresponding specialized training. Formally, for a patient i and provider j, let $d_{i,j}$ be the distance between the patient and provider, and let $\beta_{i,j}$ denote whether patient i's comorbidity and j's specialty match. For example, if patient i has a heart-related comorbidity and provider j has a cardiology specialty, then $\beta_{i,j} = 1$. The match quality is $\theta_{i,j} = \alpha + (1-\alpha)\big(\omega\beta_{i,j} + (1-\omega)(\frac{\bar{d}}{d_{i,j}} - 1)\big)$, where ω and α are hyperparameters that modulate the impact of distance and comorbidities. Here, ω weighs between comorbidities and distance, and α sets the threshold match quality for a patient at distance $d_{i,j} = \bar{d}$.

We compute $d_{i,j}$ by sampling patient locations from Connecticut zip codes and obtaining provider locations from a Medicare dataset [25]. We obtain $\beta_{i,j}$ from prior work on comorbidity rates [27] and using information on provider specialties from a Medicare dataset [25]. Finally, we set $\bar{d} = 20.2$ since prior work demonstrates the average distance threshold for patients is 20.2 miles [35]. We set the utility for not matching as $\theta_{i,M+1} = 0.25$ and let the capacity be $\bar{c}_j \sim \text{Poisson}(1)$ for simplicity.

B Other Ablations

B.1 Number of Samples

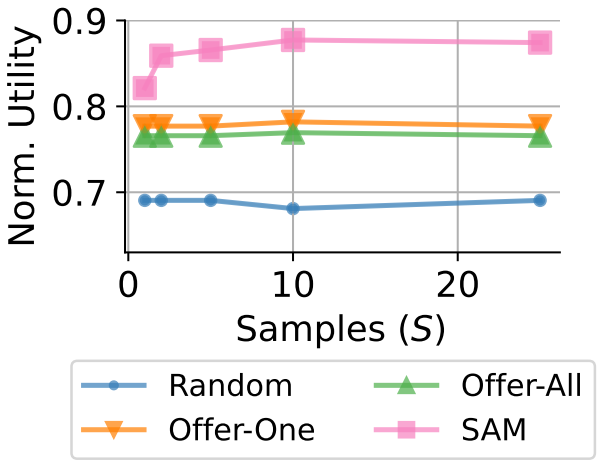


Fig. 3: We vary the number of samples S used by SAM, and find that beyond a certain threshold, performance quickly plateaus. Even $S = 10$ samples is sufficient because it can sufficiently estimate the variance in the dual variable $\hat{\lambda}$.

In Figure 3, we vary the number of samples $S \in \{1, 2, 5, 10, 25\}$ and compute the performance of SAM. We find that performance quickly plateaus with increasing S, with $S = 10$ captures 99.8% of the performance of $S = 25$. The number of samples quickly plateaus because a few samples suffice to accurately estimate $\hat{\lambda}_j$. The primary role of the samples is to better estimate $\hat{\lambda}_j$, which quickly converges in practice. While some adversarial noise distributions can lead to a larger dependence on S, in practice, we find that only a few samples are needed for good performance.

B.2 Effect of Capacity and Market Dynamics

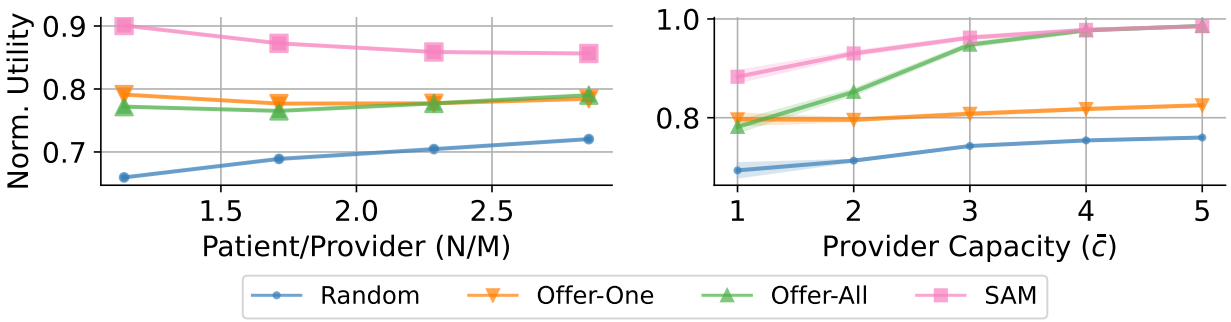


Fig. 4: Performance when varying the patient–provider ratio (left) and the average capacity per-provider (right). SAM provides the largest improvement over baseline policies when provider availability is limited, either because demand is high relative to supply or because individual provider capacities are small.

To explore congestion effects, we plot performance when varying N (while fixing $M = 700$) and varying the average provider capacity $\bar{c}$ (while fixing $N = 1225$, $M = 700$) in Figure 4. Higher patient load (increasing N) slightly reduces the performance of SAM from 0.90 to 0.86 normalized utility, with similar small changes in performance for offer-one and offer-all. More impactful is the role that provider capacity plays; when $\bar{c} = 1$, SAM is 12% better than all alternatives, but at $\bar{c} = 5$, offer-all and SAM achieve the same performance. Increasing provider capacity $\bar{c}$ reduces inter-patient competition, which allows menus such as offer-all to do well. However, increasing or decreasing N/M changes the number of patients, but still maintains inter-patient competition, requiring careful menu design.

B.3 Verifying Regularity Conditions

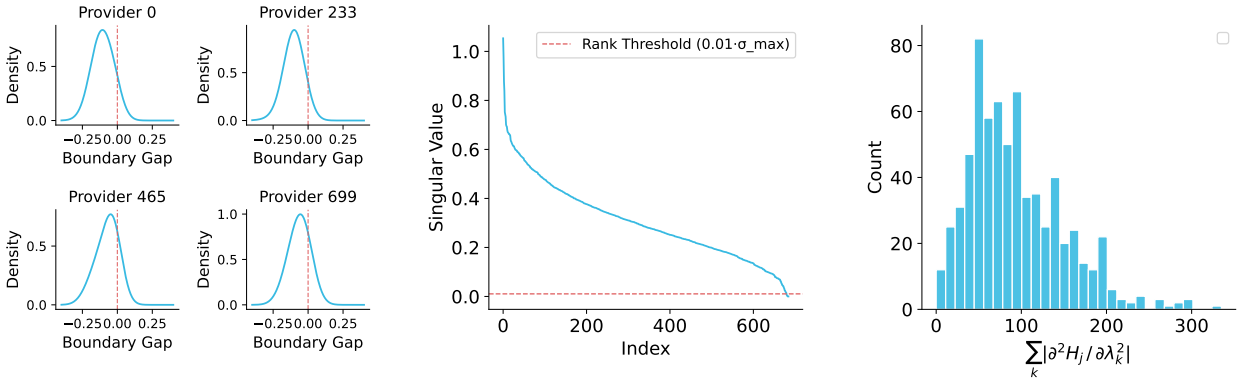


Fig. 5: (Left) Kernel density estimates of the signed boundary gap for four representative active providers. The density is smooth and continuous near zero (red dashed line) across providers, supporting the twice-differentiability condition. (Middle) Distribution of singular values for $DH(\boldsymbol{\lambda}^*)$. Almost all singular values are above 0, with the near-zero singular values corresponding to near-substitute provider pairs. (Right) Distribution of second derivatives for H_j. Most second derivatives concentrate around the median.

We verify the three regularity conditions from Section 5 empirically for our fixed population of N patients. To avoid instability, we slightly perturb $\theta + \eta$, with $\eta \sim \mathcal{N}(0, 0.01^2)$, which mimics the smoothing effect of the noise draws $\delta^{(s)}$ across scenarios. We verify twice differentiability by plotting the distribution of Δ for different providers in Figure 5 (left), and find that it is smooth and continuous near assignment boundaries for our fixed patient population. Here, the boundary gap corresponds to when $\Delta_{i,j} - \lambda_j = 0$, or when patients are indifferent about matching to a provider. For non-singularity, Figure 5 (middle) shows that 679/688 singular values are non-zero for $DH(\boldsymbol{\lambda}^*)$, with the near-zero singular values corresponding to near-substitute provider pairs that are nearly identical in the utility function for this patient population. Finally, we verify bounded second derivatives by plotting the distribution of second derivatives, and showing that these derivatives are well-concentrated across active providers, with a median of 80 and 96% of providers below 200. Together, these results confirm that the regularity conditions hold in our fixed semi-synthetic patient population for the generic case of distinguishable providers, consistent with the requirements of Theorem 3.

B.4 MNL Choice Models

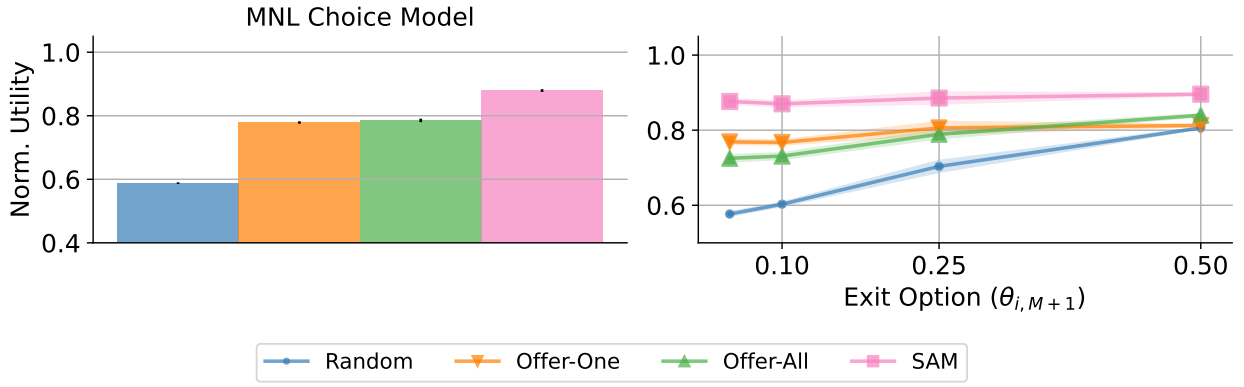


Fig. 6: Performance under the MNL choice model (left) and varied exit option (right). SAM maintains its advantage even under stochastic choice models such as MNL.

We explore the performance of SAM when patients respond according to an MNL choice model and when varying the exit option $\theta_{i,M+1}$ (Figure 6). Here, we let patients respond according to an MNL model so that responses are stochastic and dependent on the menu of options given. SAM performs best under an MNL model, and is 8% better than offer-one. SAM performs well for MNL because it offers tailored menus which restricts a patient's choice only to a curated set of high-utility options. Meanwhile, the value of the exit option has little impact on SAM's performance, though the performance of baselines improves with larger exit option because even poorly constructed menus receive a higher minimum reward.

C Deferred Proofs

C.1 Computational Intractability

Proof. Proof of Theorem 1: We prove weak NP-hardness by reduction from the *Partition* problem, which is weakly NP-complete [23]. An instance of *Partition* is a set of weights $w_1, \ldots, w_N \in \mathbb{N}$ with $w_i \geq 1$, and the goal is to find disjoint sets S_1, S_2 with $S_1 \cup S_2 = [N]$ minimizing $\left|\sum_{i \in S_1} w_i - \sum_{i \in S_2} w_i\right|$.

We reduce this problem to the problem of finding an optimal menu design, and in particular, of finding an optimal assortment $\mathbf{X}$. Given weights w_i, construct the following menu design instance. Let $M = 2$, $k = 1$,

$\varepsilon = 1$, $\bar{c}_1 = \bar{c}_2 = 1$, and $\mathcal{D}$ uniform over $[N]$. Set the exit utility $\theta_{i,3} = \gamma$ for all i, with $0 < \gamma < 1$, observed without noise. For providers $j \in \{1, 2\}$, let the true utilities be drawn independently with

$$\theta_{i,j} = \begin{cases} 1 & \text{with probability } q_i := 1 - \exp(-w_i), \\ 0 & \text{otherwise,} \end{cases}$$

so $\mathcal{P}_{\boldsymbol{\theta}}$ is the product of these Bernoulli marginals. The platform observes the unbiased estimate $\hat{\theta}_{i,j} = q_i$ for $j \in \{1, 2\}$. The induced noise $\delta_{i,j} = \hat{\theta}_{i,j} - \theta_{i,j} \in \{q_i,\, q_i - 1\}$ has mean zero,

$$\mathbb{E}[\delta_{i,j}] = q_i - \mathbb{E}[\theta_{i,j}] = q_i - q_i = 0,$$

and satisfies $\|\boldsymbol{\delta}\|_\infty = \max_i \max(q_i, 1 - q_i) \le \varepsilon = 1$, with $\theta_{i,j} \in \{0, 1\} \subseteq [0, 1]$.

Since $k = 1$, a menu assigns each patient to at most one provider and is described by disjoint sets $S_1, S_2 \subseteq [N]$, where S_j is the set offered provider j. Provider j is matched whenever some patient in S_j realizes $\theta_{i,j} = 1$, i.e., with probability $1 - \exp(-\sum_{i \in S_j} w_i)$, which is increasing in S_j. As a match yields utility $1 > \gamma$, offering every patient is optimal, so we take $S_1 \cup S_2 = [N]$ and $\{S_1, S_2\}$ partitions $[N]$.

Let $W_j := \sum_{i \in S_j} w_i$, so $W_1 + W_2 = W := \sum_i w_i$ is fixed. A matched patient yields utility 1 and an unmatched patient yields γ, and the expected number of matches is $(1 - \exp(-W_1)) + (1 - \exp(-W_2))$, so the expected total utility (independent of arrival order) is

$$(1 - \gamma)\big[\, 2 - \exp(-W_1) - \exp(-W_2) \,\big] + N\gamma.$$

Since $1 - \gamma > 0$ and $N\gamma$ is constant, this is maximized by minimizing $\exp(-W_1) + \exp(-W_2)$ at fixed $W_1 + W_2$, which by convexity occurs at the partition minimizing $|W_1 - W_2|$. Solving menu optimization thus solves the *Partition* problem, so the problem is weakly NP-hard, even under $k = 1$, $M = 2$, unit capacities, and zero-mean Bernoulli noise.

We note that the construction uses zero-mean noise and the unbiased estimate $\hat{\theta}_{i,j} = q_i$, so the platform effectively knows the demand means exactly; hardness nonetheless holds, since the *Partition* problem is NP-hard even with known weights. The difficulty is thus intrinsic to balancing expected demand against shared capacity, rather than an artifact of biased or adversarial noise, and it persists in the same zero-mean regime under which we establish the guarantees for SAM in Section 5.

Proof. Proof of Theorem 2:

Instance Construction. Let $N = \alpha \sum_{j=1}^{M} \bar{c}_j$. Partition the N patients into M disjoint groups $G_1, \ldots, G_M$ with $|G_j| = \alpha \bar{c}_j$.

Fix a small constant γ with $0 < \gamma \ll \frac{1}{2} - \varepsilon$, and set the exit utility $\theta_{i,M+1} = \gamma$ for all patients i. Set $\theta_{i,j} = 0$ for all $i \notin G_j$; patients outside group j receive zero utility from provider j and will always prefer the exit option over provider j, since $0 < \gamma$. Set

$$\hat{\theta}_{i,j} = \frac{1}{2} \quad \text{for all } i \in G_j, \quad \hat{\theta}_{i,j} = 0 \quad \text{for all } i \notin G_j.$$

Additionally, let $\hat{\theta}_{i,M+1} = \gamma$. Next, let

$$\delta_{i,j} = \begin{cases} +\varepsilon & \text{with probability } \frac{1}{2} \\ -\varepsilon & \text{with probability } \frac{1}{2} \end{cases} \quad \text{for all } i \in G_j.$$

We note that, $\delta_{i,j} = \hat{\theta}_{i,j} - \theta_{i,j} \in [-\varepsilon, \varepsilon]$ and is zero mean.

Omniscient welfare. The optimal omniscient algorithm (with knowledge of θ) offers provider j to the top-$\bar{c}_j$ patients with the smallest values of $\delta_{i,j}$. This is equivalent to

$$OPT = \sum_{j=1}^{M} \left[\bar{c}_j \cdot \frac{1}{2} - \sum_{i \in \text{top-}\bar{c}_j(G_j)} \delta_{i,j} \right] = \frac{1}{2} \sum_{j=1}^{M} \bar{c}_j \; - \sum_{j=1}^{M} \sum_{i \in \text{top-}\bar{c}_j(G_j)} \delta_{i,j},$$

where top-$\bar{c}_j(G_j)$ denotes the $\bar{c}_j$ patients in G_j with the smallest $\delta_{i,j}$ (since $\theta_{i,j} = 0$ is constant within G_j, ranking by $\hat{\theta}_{i,j} + \delta_{i,j}$ within the group is the same as ranking by $\delta_{i,j}$ alone).

The values $\{\delta_{i,j}\}_{i \in G_j}$ are $|G_j| = \alpha \bar{c}_j$ i.i.d. $\pm\varepsilon$ coin flips. Let X_j denote the number of $i \in G_j$ with $\delta_{i,j} = -\varepsilon$; then $X_j \sim \text{Binomial}(\alpha \bar{c}_j, \frac{1}{2})$, with mean $\frac{\alpha \bar{c}_j}{2}$. By a Chernoff bound,

$$\Pr\left[X_j < \bar{c}_j\right] \;\le\; \exp\left(-c\,\alpha \bar{c}_j\right)$$

for a constant $c > 0$, provided $\alpha > 2$ (so that the threshold $\bar{c}_j$ is a constant fraction below the mean $\alpha \bar{c}_j/2$). Hence for any fixed $\alpha > 2$, as $\bar{c}_j$ grows (or via a union bound over $j = 1, \ldots, M$ when M is polynomial in the instance size), with probability $1 - o(1)$ we have $X_j \ge \bar{c}_j$ simultaneously for all j, i.e. each group has at least $\bar{c}_j$ patients with $\delta_{i,j} = -\varepsilon$. On this event, the top-$\bar{c}_j$ patients in G_j by $\delta_{i,j}$ all have $\delta_{i,j} = -\varepsilon$, so

$$\sum_{i \in \text{top-}\bar{c}_j(G_j)} \delta_{i,j} \;=\; -\bar{c}_j\,\varepsilon.$$

Combining, with probability $1 - o(1)$,

$$OPT \;=\; \left(\frac{1}{2} + \varepsilon\right) \sum_{j=1}^{M} \bar{c}_j,$$

and in particular $\mathbb{E}[OPT] \to \left(\frac{1}{2} + \varepsilon\right) \sum_j \bar{c}_j$ as $\alpha \to \infty$.

Algorithm welfare and the gap. Fix provider j. Any patient outside G_j has zero utility for provider j and never consumes a slot, so we may take $S_j := \{i : X_{i,j} = 1\} \subseteq G_j$. In expectation, any subset of S_j gets utility $\frac{1}{2}\bar{c}_j$. The total algorithm welfare is then $\frac{1}{2}\sum_j \bar{c}_j$. Taking the difference with OPT gives that the gap is $\varepsilon \sum_j \bar{c}_j$.

A similar argument extends to continuous noise satisfying regularity condition R1, such as $\delta_{i,j}$ being drawn from a truncated Gaussian or Laplace distributions, where we replace the Chernoff bound with a quantile concentration argument.

C.2 Offer-One Guarantee

Proof. Proof of Proposition 3: We first construct a worst case instance. Suppose N and ε are given, and let $M = N$. For simplicity, assume that N is divisible by two. Next, for odd i, let $\hat{\theta}_{i,i} = \hat{\theta}_{i+1,i+1} = \frac{1}{2} + \gamma$ and $\hat{\theta}_{i+1,i} = \hat{\theta}_{i,i+1} = \frac{1}{2}$. Let $\hat{\theta}_{i,j} = 0$ otherwise. Let $\bar{c}_j = 1$, so that the offer-one algorithm pairs each patient i with provider i; that is $X_{i,i} = 1$ and $X_{i,j} = 0$ for $i \neq j$. Next, let $\theta_{i,i} = \theta_{i+1,i+1} = \frac{1}{2} - \varepsilon + \gamma$ and $\theta_{i,i+1} = \theta_{i+1,i} = \frac{1}{2} + \varepsilon$ with probability $\frac{1}{2}$. Let $\theta_{i,i} = \theta_{i+1,i+1} = \frac{1}{2} + \varepsilon + \gamma$ and $\theta_{i,i+1} = \theta_{i+1,i} = \frac{1}{2} - \varepsilon$ otherwise. Then $\text{OPT} = \frac{1}{2}N(\frac{1}{2}+\varepsilon+\gamma) + \frac{1}{2}N(\frac{1}{2}+\varepsilon) = \frac{N}{2} + N\varepsilon + \frac{N\gamma}{2}$, while $\mathbb{E}[V_\theta(\mathbf{X}^{\text{one}})] = \frac{1}{2}N(\frac{1}{2}+\varepsilon+\gamma) + \frac{1}{2}N(\frac{1}{2}-\varepsilon+\gamma) = \frac{N}{2} + \gamma N$. Then we have that

$$\mathbb{E}[\text{OPT} - V_\theta(\mathbf{X}^{\text{one}})] = \varepsilon N - \frac{\gamma N}{2}.$$

Letting $\gamma \to 0$ shows that $\mathbb{E}[\text{OPT} - V_\theta(\mathbf{X}^{\text{one}})] = \varepsilon N$.

Next, we show that such a bound is tight. Since each patient selects at most one provider and utilities are bounded in $[0, 1]$, the total contribution of each patient to the objective differs by at most ε under entrywise perturbations. For a fixed menu $\mathbf{X}$ with $\|\mathbf{X}_{*,j}\|_1 \le 1$, we note that

$$|V_\theta(\mathbf{X}) - V_{\hat{\theta}}(\mathbf{X})| = |\mathbb{E}[\sum_{i=1}^{N} \theta_{i,s_i(\mathbf{X},\boldsymbol{\sigma},\theta)} - \sum_{i=1}^{N} \hat{\theta}_{i,s_i(\mathbf{X},\boldsymbol{\sigma},\hat{\theta})}]| \le \varepsilon N.$$

By Section 4.1, one-provider menus are optimal under known utilities, so both $\mathbf{X}^*$ and $\mathbf{X}^{\text{one}}$ are matchings; each therefore induces a fixed assignment, and the bound above holds entrywise. Next, let $\mathbf{X}^* = \arg\max_{\mathbf{X}} V_\theta(\mathbf{X})$ and $\mathbf{X}^{\text{one}} = \arg\max_{\mathbf{X}} V_{\hat{\theta}}(\mathbf{X})$. Then we have

$$\mathbb{E}[V_\theta(\mathbf{X}^*) - V_\theta(\mathbf{X}^{\text{one}})] = \mathbb{E}[V_\theta(\mathbf{X}^*) - V_{\hat{\theta}}(\mathbf{X}^*)] + \mathbb{E}[V_{\hat{\theta}}(\mathbf{X}^*) - V_{\hat{\theta}}(\mathbf{X}^{\text{one}})] + \mathbb{E}[V_{\hat{\theta}}(\mathbf{X}^{\text{one}}) - V_\theta(\mathbf{X}^{\text{one}})] \le \varepsilon N.$$

Here, the first is bounded by εN, while the second term is non-positive due to the optimality of $\mathbf{X}^{\text{one}}$ on $\hat{\theta}$. For the third term, recall that $\mathbf{X}^{\text{one}}$ depends only on the observed $\hat{\theta}$ and is therefore fixed, so $V_{\hat{\theta}}(\mathbf{X}^{\text{one}})$ is deterministic. In $\mathbf{X}^{\text{one}}$, each patient i offered its single provider $j(i)$ realizes utility $\max\{\theta_{i,j(i)},\, \theta_{i,M+1}\}$, a convex function of θ. By Jensen's inequality together with $\mathbb{E}[\theta] = \hat{\theta}$,

$$\mathbb{E}\big[V_\theta(\mathbf{X}^{\text{one}})\big] = \sum_{i=1}^{N} \mathbb{E}\big[\max\{\theta_{i,j(i)},\, \theta_{i,M+1}\}\big] \;\ge\; \sum_{i=1}^{N} \max\{\hat{\theta}_{i,j(i)},\, \hat{\theta}_{i,M+1}\} = V_{\hat{\theta}}(\mathbf{X}^{\text{one}}),$$

so the third term $\mathbb{E}[V_{\hat{\theta}}(\mathbf{X}^{\text{one}}) - V_\theta(\mathbf{X}^{\text{one}})] \le 0$, with equality when no patient is near the exit threshold (as in the instance constructed above). Combining the three terms yields $\mathbb{E}[\text{OPT} - V_\theta(\mathbf{X}^{\text{one}})] \le \varepsilon N$.

C.3 SAM Performance Guarantees

Proof. Proof of Lemma 2: We proceed in three steps.

Step 1. $\hat{H}_j(\boldsymbol{\lambda}^*) = \rho_j + O(\varepsilon^2)$. Under noisy utilities $\hat{\theta}_{i,j} = \theta_{i,j} + \delta_{i,j}$, the observed gap for patient i is $\hat{\Delta}_{i,j} = \Delta_{i,j} + (\delta_{i,j} - \delta_{i,M+1})$. Each scenario s draws an independent resample $\delta_{i,j}^{(s)}$ from the same distribution, so the effective per-scenario gap is $\hat{\Delta}_{i,j}^{(s)} = \Delta_{i,j} + \xi_{i,j}^{(s)}$, where

$$\xi_{i,j}^{(s)} := (\delta_{i,j} - \delta_{i,M+1}) + (\delta_{i,j}^{(s)} - \delta_{i,M+1}^{(s)})$$

satisfies $\mathbb{E}[\xi_{i,j}^{(s)}] = 0$, with both layers independent. The noisy assignment probability under two-layer noise is

$$\hat{H}_j(\boldsymbol{\lambda}) := \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_{\delta^{(s)}} \left[\mathbf{1} \left[\Delta_{i,j} + \xi_{i,j} - \lambda_j = \max\left(0,\, \max_{j' \in [M]} (\Delta_{i,j'} + \xi_{i,j'} - \lambda_{j'})\right)\right]\right]$$

Adding $\xi_{i,j'}^{(s)}$ to $\Delta_{i,j'}$ is equivalent to shifting each threshold by $-\xi_{i,j'}^{(s)}$, so conditioning on the noise vector $\boldsymbol{\xi}_i^{(s)} := (\xi_{i,1}^{(s)}, \ldots, \xi_{i,M}^{(s)})^\top$ gives

$$\hat{H}_j(\boldsymbol{\lambda}^*) = \mathbb{E}_{\boldsymbol{\xi}_i^{(s)}} \left[H_j(\boldsymbol{\lambda}^* - \boldsymbol{\xi}_i^{(s)}) \right]. \tag{3}$$

By R1 the empirical share H_j is twice differentiable near the assignment boundary (the noise density smooths the otherwise piecewise-constant share), so we can Taylor expand H_j at $\boldsymbol{\lambda}^*$ and take expectations:

$$\hat{H}_j(\boldsymbol{\lambda}^*) = H_j(\boldsymbol{\lambda}^*) - \nabla_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^*)^\top \underbrace{\mathbb{E}[\boldsymbol{\xi}_i^{(s)}]}_{=\mathbf{0}} + \tfrac{1}{2} \operatorname{tr}\Big(D^2_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^*)\, \boldsymbol{\Sigma} \Big) + O(\varepsilon^3),$$

where $\boldsymbol{\Sigma} := \mathbb{E}[\boldsymbol{\xi}_i^{(s)} (\boldsymbol{\xi}_i^{(s)})^\top] \in \mathbb{R}^{M \times M}$ is the noise covariance. Since $\varepsilon \le 1$, the $O(\varepsilon^3)$ remainder is absorbed into the $O(\varepsilon^2)$ bias term under bounded third derivatives. The first-order term vanishes by mean-zero noise.

Since all noise terms are bounded by ε, every entry of $\boldsymbol{\Sigma}$ satisfies $|\Sigma_{jj'}| \le 4\varepsilon^2$ for all $j, j' \in [M]$; note that off-diagonal entries are nonzero because the shared exit-option noise $\delta_{i,M+1}$ appears in every $\xi_{i,j}$. Therefore

$$\tfrac{1}{2}\operatorname{tr}\Big(D^2_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^*)\,\boldsymbol{\Sigma}\Big) = \tfrac{1}{2}\sum_{k=1}^{M}\sum_{k'=1}^{M}\frac{\partial^2 H_j(\boldsymbol{\lambda}^*)}{\partial\lambda_k\partial\lambda_{k'}}\,\Sigma_{kk'} \le 2\varepsilon^2\sum_{k=1}^{M}\sum_{k'=1}^{M}\left|\frac{\partial^2 H_j(\boldsymbol{\lambda}^*)}{\partial\lambda_k\partial\lambda_{k'}}\right| = O(\varepsilon^2)$$

by R3. Combined with market clearing $H_j(\boldsymbol{\lambda}^*) = \rho_j$ for $j \in J^*$:

$$\hat{H}_j(\boldsymbol{\lambda}^*) = H_j(\boldsymbol{\lambda}^*) + O(\varepsilon^2) = \rho_j + O(\varepsilon^2) \qquad \forall\, j \in J^*. \tag{4}$$

For $j \notin J^*$, complementary slackness gives $H_j(\boldsymbol{\lambda}^*) \le \rho_j$; let $\gamma_j := \rho_j - H_j(\boldsymbol{\lambda}^*) \ge 0$. Let $\hat{H}_j^{(s)}(\boldsymbol{\lambda}^*)$ denote the realized fraction of patients selecting j in scenario s at price $\boldsymbol{\lambda}^*$, and set $\zeta_{j,s} := \hat{H}_j^{(s)}(\boldsymbol{\lambda}^*) - H_j(\boldsymbol{\lambda}^*)$. Since $\mathbb{E}[\zeta_{j,s}] = O(\varepsilon^2)$ (from step 1) and $\gamma_j \ge 0$,

$$\mathbb{E}[(\zeta_{j,s} - \gamma_j)_+] \le \mathbb{E}[(\zeta_{j,s})_+] \le \mathbb{E}[|\zeta_{j,s}|] \le |\mathbb{E}[\zeta_{j,s}]| + \sqrt{\operatorname{Var}(\zeta_{j,s})} = O(\varepsilon^2) + O(1/\sqrt{N}),$$

and $\operatorname{Var}(\zeta_{j,s}) = O(1/N)$ since $\hat{H}_j^{(s)}(\boldsymbol{\lambda}^*)$ averages N bounded indicators.

Provider j binds in scenario s ($\lambda_{j,s} > 0$) only if $\zeta_{j,s} > \gamma_j$. Otherwise, $\lambda_{j,s} = \lambda_j^* = 0$. When provider j binds in scenario s (i.e., $\zeta_{j,s} > \gamma_j$), the per-scenario dual $\lambda_{j,s}$ is proportional to the excess demand $(\zeta_{j,s} - \gamma_j)_+$ by an implicit function theorem argument (formalized in Step 2), so in both cases:

$$\lambda_{j,s} \le \kappa(\zeta_{j,s} - \gamma_j)_+.$$

Since $\mathbb{E}[(\zeta_{j,s} - \gamma_j)_+] = O(\varepsilon^2)$ (as $\mathbb{E}[\zeta_{j,s}] = O(\varepsilon^2)$ by Step 1 and $\gamma_j \ge 0$) and $(\zeta_{j,s} - \gamma_j)_+ \in [0,1]$ is independent across scenarios, averaging over S scenarios and applying Hoeffding's inequality to $\frac{1}{S}\sum_s \lambda_{j,s} \le \frac{\kappa}{S}\sum_s(\zeta_{j,s} - \gamma_j)_+$ gives, with probability at least $1 - \nu/M$:

$$\hat{\lambda}_j^{\mathrm{SAM}} \le O(\varepsilon^2) + O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right) = \lambda_j^* + O(\varepsilon^2) + O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right)$$

uniformly over $j \notin J^*$.

Step 2. Implicit Function Theorem gives $\|\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\|_\infty = O(\varepsilon^2)$ for the fixed point $\hat{\boldsymbol{\lambda}}$ of $\hat{H}$. Differentiating $\hat{H}_j(\boldsymbol{\lambda}^*) = \mathbb{E}_{\boldsymbol{\xi}_i^{(s)}}[H_j(\boldsymbol{\lambda}^* - \boldsymbol{\xi}_i^{(s)})]$ with respect to $\boldsymbol{\lambda}$ gives

$$D_{\boldsymbol{\lambda}}\hat{H}_j(\boldsymbol{\lambda}^*) = \mathbb{E}_{\boldsymbol{\xi}_i^{(s)}}\left[D_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^* - \boldsymbol{\xi}_i^{(s)})\right].$$

Applying a first-order Taylor expansion to $D_{\boldsymbol{\lambda}} H_j$ at $\boldsymbol{\lambda}^*$:

$$\mathbb{E}\Big[D_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^* - \boldsymbol{\xi}_i^{(s)})\Big] = D_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^*) - D^2_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^*)\,\mathbb{E}[\boldsymbol{\xi}_i^{(s)}] + O(\|\boldsymbol{\xi}_i^{(s)}\|^2).$$

Since $\mathbb{E}[\boldsymbol{\xi}_i^{(s)}] = \mathbf{0}$ and third derivatives are bounded, by R3:

$$D_{\boldsymbol{\lambda}}\hat{H}_j(\boldsymbol{\lambda}^*) = D_{\boldsymbol{\lambda}} H_j(\boldsymbol{\lambda}^*) + O(\varepsilon^2).$$

Stacking over $j \in J^*$ gives $D_{\boldsymbol{\lambda}}\hat{H}(\boldsymbol{\lambda}^*)|_{J^*} = D_{\boldsymbol{\lambda}} H(\boldsymbol{\lambda}^*)|_{J^*} + O(\varepsilon^2)$ as matrices in $\mathbb{R}^{|J^*|\times|J^*|}$.

The implicit function theorem states that if $F(\boldsymbol{\lambda}) = \boldsymbol{\rho}$ has solution $\boldsymbol{\lambda}^*$ and $D_{\boldsymbol{\lambda}} F(\boldsymbol{\lambda}^*)$ is non-singular, then for any perturbed system $\tilde{F}(\boldsymbol{\lambda}) = \boldsymbol{\rho}$ with $\tilde{F}$ smooth near $\boldsymbol{\lambda}^*$, the perturbed solution $\hat{\boldsymbol{\lambda}}$ satisfies

$$\|\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\| \;\le\; \|[D_{\boldsymbol{\lambda}}\tilde{F}(\boldsymbol{\lambda}^*)]^{-1}\| \cdot \|\tilde{F}(\boldsymbol{\lambda}^*) - \boldsymbol{\rho}\| \;+\; O(\|\tilde{F}(\boldsymbol{\lambda}^*) - \boldsymbol{\rho}\|^2).$$

We apply this with $F = H|_{J^*}$ and $\tilde{F} = \hat{H}|_{J^*}$. The Jacobian bound $\|[D_{\boldsymbol{\lambda}}\hat{H}(\boldsymbol{\lambda}^*)|_{J^*}]^{-1}\| \le \kappa$ follows from R2 together with $D_{\boldsymbol{\lambda}}\hat{H}(\boldsymbol{\lambda}^*)|_{J^*} = D_{\boldsymbol{\lambda}} H(\boldsymbol{\lambda}^*)|_{J^*} + O(\varepsilon^2)$, which ensures the Jacobian of $\hat{H}|_{J^*}$ remains non-singular with inverse norm bounded by κ for small enough ε. Substituting $\|\hat{H}(\boldsymbol{\lambda}^*)|_{J^*} - \boldsymbol{\rho}\| = O(\varepsilon^2)$ from Step 1 gives:

$$\|\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}^*\| \;\le\; \kappa \cdot O(\varepsilon^2) + O(\varepsilon^4) \;=\; O(\varepsilon^2).$$

This establishes the bound for the fixed point $\hat{\boldsymbol{\lambda}}$ satisfying $\hat{H}(\hat{\boldsymbol{\lambda}})|_{J^*} = \boldsymbol{\rho}$.

Step 3. Finite-sample concentration gives $\|\hat{\boldsymbol{\lambda}}^{\mathrm{SAM}} - \hat{\boldsymbol{\lambda}}\|_\infty = O\Big(\sqrt{\log(M/\nu)/S}\Big)$. Fix $j \in J^*$ and scenario s. The per-scenario dual $\lambda_{j,s}$ satisfies the empirical market-clearing condition $\hat{H}^{\mathrm{emp}}_{j,s}(\lambda_{j,s}) = \rho_j$, while the fixed point satisfies $\hat{H}_j(\hat{\lambda}_j) = \rho_j$. Here, this differs from $\mathbb{E}[\lambda_{j,s}]$, the expected per-scenario dual, by $O(1/\sqrt{N})$, since solving and averaging do not commute; the gap closes as $N \to \infty$.

Applying the implicit function theorem as in Step 2 linearizes the difference:

$$\big|\lambda_{j,s} - \hat{\lambda}_j\big| \;\le\; \kappa \cdot \big|\hat{H}^{\mathrm{emp}}_{j,s}(\hat{\lambda}_j) - \rho_j\big| + O\Big(\big|\hat{H}^{\mathrm{emp}}_{j,s}(\hat{\lambda}_j) - \rho_j\big|^2\Big).$$

We can view $\hat{H}^{\mathrm{emp}}_{j,s}(\hat{\lambda}_j)$ as an empirical mean of N bounded indicators, with each patient impacting the average by at most $\frac{1}{N}$ for a given provider j and scenario s. Since $\hat{\lambda}_j$ is the fixed point of $\hat{H}_j$, we have $\mathbb{E}[\hat{H}^{\mathrm{emp}}_{j,s}(\hat{\lambda}_j)] = \hat{H}_j(\hat{\lambda}_j) = \rho_j$. By McDiarmid's inequality applied around this mean, $|\hat{H}^{\mathrm{emp}}_{j,s}(\hat{\lambda}_j) - \rho_j| \le O(1/\sqrt{N})$, so the IFT remainder is $O(1/N)$ and:

$$\big|\lambda_{j,s} - \hat{\lambda}_j\big| \le O\Big(\frac{1}{\sqrt{N}}\Big).$$

Since $\lambda_{j,s} \ge 0$ always, each per-scenario dual is bounded in $[0, \hat{\lambda}_j + O(1/\sqrt{N})]$, and $\hat{\lambda}_j$ is bounded because it is the dual variable for a bounded LP. Since each scenario s draws an independent noise vector $\delta^{(s)}$, the S duals $\lambda_{j,1}, \ldots, \lambda_{j,S}$ are independent. Applying Hoeffding's inequality over S independent bounded random variables:

$$\left|\frac{1}{S}\sum_{s=1}^{S} \lambda_{j,s} - \mathbb{E}[\lambda_{j,s}]\right| \le O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right)$$

with probability at least $1 - \nu/M$. It remains to bound $|\mathbb{E}[\lambda_{j,s}] - \lambda^*_j|$. By the triangle inequality:

$$|\mathbb{E}[\lambda_{j,s}] - \lambda^*_j| \le |\mathbb{E}[\lambda_{j,s}] - \hat{\lambda}_j| + |\hat{\lambda}_j - \lambda^*_j| = O(1/\sqrt{N}) + O(\varepsilon^2),$$

where the first term is a finite-sample bias from solving empirical rather than population market clearing, and the second follows from Step 2.

Taking a union bound over $j \in J^*$ with $|J^*| \le M$, adding in the $1/\sqrt{N}$ term from $j \notin J^*$, and combining via the triangle inequality:

$$\|\hat{\boldsymbol{\lambda}}^{\mathrm{SAM}} - \boldsymbol{\lambda}^*\|_\infty \le \|\hat{\boldsymbol{\lambda}}^{\mathrm{SAM}} - \mathbb{E}[\lambda_{\cdot,s}]\|_\infty + \|\mathbb{E}[\lambda_{\cdot,s}] - \lambda^*\|_\infty = O(\varepsilon^2) + O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right) + O(1/\sqrt{N}).$$

Proof. Proof of Theorem 3:

Step 1. OPT *via LP strong duality.* The LP over the fixed N patients is:

$$\max_{x_{i,j} \ge 0} \sum_{i,j} \Delta_{i,j}\, x_{i,j} \quad \text{s.t.} \quad \sum_j x_{i,j} \le 1 \;[\text{dual } \mu^*_i \ge 0], \qquad \sum_i x_{i,j} \le \bar{c}_j \;[\text{dual } \lambda^*_j \ge 0]. \tag{5}$$

By LP strong duality, $\mathrm{OPT} = \sum_i \mu^*_i + \sum_j \bar{c}_j \lambda^*_j$. LP stationarity gives $\mu^*_i = \max(0, \max_j(\Delta_{i,j} - \lambda^*_j))$. This holds regardless of which constraints bind: for $j \notin J^*$, $\lambda^*_j = 0$ and the corresponding term vanishes, so the duality computation is unaffected by whether every provider is active. Therefore:

$$\mathrm{OPT} = \sum_{i=1}^{N} \theta_{i,M+1} + \sum_{i=1}^{N} \max\Big(0,\; \max_{j \in [M]} (\Delta_{i,j} - \lambda^*_j)\Big) + \sum_j \bar{c}_j \lambda^*_j. \tag{1}$$

Step 2. Dual concentration and reduction to the threshold policy. Dual concentration. By Lemma 2, with probability $\geq 1-\nu$ over the S sampled scenarios,

$$\tilde{\varepsilon} \;:=\; \|\hat{\boldsymbol{\lambda}}^{\mathrm{SAM}} - \boldsymbol{\lambda}^*\|_\infty \;\leq\; C_0\varepsilon^2 + C_1\sqrt{\frac{\log(M/\nu)}{S}} + C_2\frac{1}{\sqrt{N}}. \tag{2}$$

Define the effective noise band $\bar{\varepsilon} := 2\varepsilon + \tilde{\varepsilon}$, which captures the total uncertainty a patient-provider pair faces.

Reduction to the threshold policy. Let $\mathbf{X}^{\hat{\Delta},\hat{\lambda}}$ denote the *threshold policy* and follows the rule: include provider j in patient i's menu iff $\hat{\Delta}_{i,j} \geq \hat{\lambda}_j$, with no cardinality cap. Algorithm 1 produces the *max-form top-k policy*: include the k providers with largest positive scores $m_{i,j} = \frac{1}{S}\sum_{s=1}^{S}\max\{\Delta_{i,j}^{(s)} - \hat{\lambda}_j,\, 0\}$.

For safe-in pairs ($\Delta_{i,j} > \lambda_j^* + \bar{\varepsilon}$): $\hat{\Delta}_{i,j} \geq \Delta_{i,j} - 2\varepsilon > \hat{\lambda}_j$, so the threshold policy includes them, and their score $m_{i,j} \geq \Delta_{i,j} - \hat{\lambda}_j - O(\varepsilon) > 0$ clears the positive-score criterion of Algorithm 1. The top-k cap may still drop a safe-in pair when patient i has more than k of them, but since the cap retains the patient's highest-scoring provider for any $k \geq 1$, it removes only lower-ranked backups. For safe-out pairs ($\Delta_{i,j} < \lambda_j^* - \bar{\varepsilon}$): $\hat{\Delta}_{i,j} \leq \Delta_{i,j} + 2\varepsilon < \hat{\lambda}_j$, so the threshold policy excludes them, and $m_{i,j} \approx 0$ keeps them out of Algorithm 1's top-k. The patient's selected provider therefore differs between the two policies only for near-threshold pairs ($|\Delta_{i,j} - \lambda_j^*| \leq \bar{\varepsilon}$), where each discrepant provider contributes at most $O(\bar{\varepsilon})$ surplus to patient i. The total welfare cost of these discrepancies, together with the top-k cap, is $\varepsilon_k := O(\varepsilon\sum_j \bar{c}_j)$, bounded in Step 3.

Hence:

$$\mathrm{OPT} - R(\mathbf{X}^{\mathrm{SAM}}) \;\leq\; \big[\mathrm{OPT} - R(\mathbf{X}^{\hat{\Delta},\hat{\lambda}})\big] + \varepsilon_k.$$

Step 3. The top-k constraint contributes $O\Big(\varepsilon\sum_j \bar{c}_j\Big)$. When there is zero noise, patient i's LP welfare is $\mu_i^*(X_i) = \max_{j\in X_i}(\Delta_{i,j} - \lambda_j^*)^+$, determined solely by the highest-scoring provider in X_i. Let $j^* \in \arg\max_j(\Delta_{i,j} - \lambda_j^*)^+$ be any maximizer, with ties broken arbitrarily, so the top-k policy retains $j^*(i)$ for any $k \geq 1$ and achieves the same LP welfare as the uncapped policy.

With noise $\varepsilon > 0$, the top-k policy diverges from the uncapped policy only if some provider $j' \neq j^*(i)$ achieves a higher noisy score than $j^*(i)$, displacing all maximizers from the top-k. This requires

$$(\Delta_{i,j'} - \lambda_{j'}^*) + O(\varepsilon) > (\Delta_{i,j^*(i)} - \lambda_{j^*(i)}^*) - O(\varepsilon),$$

so j' must lie within $O(\varepsilon)$ of $j^*(i)$ in true score. When displacement occurs, the patient receives j' instead of $j^*(i)$, losing at most $(\Delta_{i,j^*(i)} - \lambda_{j^*(i)}^*) - (\Delta_{i,j'} - \lambda_{j'}^*) = O(\varepsilon)$ in welfare.

Counting affected patients. A patient with $\max_j(\Delta_{i,j} - \lambda_j^*) \leq 0$ exits under any menu and suffers no truncation loss. Only patients with a positive argmax are at risk. Since the events $\{j^*(i) = j\}$ are mutually exclusive,

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}[\max_j(\Delta_{i,j} - \lambda_j^*) > 0] = \sum_j H_j(\boldsymbol{\lambda}^*) \leq \sum_j \rho_j = \frac{\sum_j \bar{c}_j}{N},$$

so at most $\sum_j \bar{c}_j$ patients have a positive argmax. Each loses at most $O(\varepsilon)$, giving total truncation cost:

$$\varepsilon_k \;\leq\; \sum_j \bar{c}_j \cdot O(\varepsilon) \;=\; O\left(\varepsilon\sum_j \bar{c}_j\right).$$

Step 4. Welfare error per near-threshold pair is $O(\varepsilon)$. We next focus on bounding $\big[\mathrm{OPT} - R(\mathbf{X}^{\hat{\Delta},\hat{\lambda}})\big] = \big[R(\mathbf{X}^{\Delta,\lambda^*}) - R(\mathbf{X}^{\hat{\Delta},\hat{\lambda}})\big]$. Fix a near-threshold pair (i,j), meaning $|\Delta_{i,j} - \lambda_j^*| \leq \bar{\varepsilon} = O(\varepsilon)$.

False negative ($\mathbf{X}^{\Delta,\lambda^*}$ includes (i,j), $\mathbf{X}^{\hat{\Delta},\hat{\lambda}}$ excludes it): Provider j is in patient i's exact threshold menu with $\Delta_{i,j} - \lambda_j^* \in (0, \bar{\varepsilon}]$, so excluding it costs at most $\bar{\varepsilon} = O(\varepsilon)$.

False positive ($\mathbf{X}^{\Delta,\lambda^*}$ excludes (i,j), $\mathbf{X}^{\hat{\Delta},\hat{\lambda}}$ includes it): Patient i is assigned to j even though $\Delta_{i,j} < \lambda_j^*$. By LP duality, λ_j^* is the shadow price of one unit of capacity at provider j, so the welfare loss from

allocating that unit to patient i rather than optimally is exactly $\lambda_j^* - \Delta_{i,j}$. Since (i,j) is near-threshold, $\lambda_j^* - \Delta_{i,j} \le \bar{\varepsilon} = O(\varepsilon)$.

In both cases the per-pair welfare error is $O(\varepsilon)$.

Step 5. Total misclassification error is $O\Big(\varepsilon \sum_j \bar{c}_j\Big)$. Fix provider j. A false negative at (i,j) incurs welfare loss only when j is the unique maximizer assigned to patient i; otherwise patient i still selects a maximizer from both menus and excluding j from the noisy menu costs nothing. The welfare-relevant false negatives are therefore the subset $\{i : j \in \arg\max_{j'}(\Delta_{i,j'} - \lambda_{j'})\}$, which by market clearing has at most $\bar{c}_j$ members. Provider j has capacity $\bar{c}_j$, so at most $\bar{c}_j$ patients can be false positives. Moreover, both false negatives and false positives can only be near-threshold pairs: a patient with $\Delta_{i,j} - \lambda_j^* > \bar{\varepsilon}$ will have $\hat{\Delta}_{i,j} - \hat{\lambda}_j > 0$ (safe-in, Step 2), and a patient with $\Delta_{i,j} - \lambda_j^* < -\bar{\varepsilon}$ will have $\hat{\Delta}_{i,j} - \hat{\lambda}_j < 0$ (safe-out, Step 2). Hence misclassifications are a subset of the at-most-$\bar{c}_j$ capacity-limited events, each contributing $O(\varepsilon)$ by Step 4:

$$\mathrm{Error}_j \;=\; O(\varepsilon \bar{c}_j).$$

Summing over j:

$$\mathrm{OPT} - R(\mathbf{X}^{\hat{\Delta},\hat{\lambda}}) \;=\; O\left(\varepsilon \sum_j \bar{c}_j\right). \tag{3}$$

Step 6. Finite-S correction via Lemma 2. Steps 2–5 used $\tilde{\varepsilon} = O(\varepsilon^2)$, which corresponds to the large-S limit of Lemma 2. For finite S, Lemma 2 gives the sharper, general bound: with probability at least $1-\nu$ over the S sampled scenarios,

$$\tilde{\varepsilon} \;:=\; \|\hat{\boldsymbol{\lambda}}^{\mathrm{SAM}} - \boldsymbol{\lambda}^*\|_\infty \;\le\; O(\varepsilon^2) + O\left(\sqrt{\frac{\log(M/\nu)}{S}}\right) + O(1/\sqrt{N}).$$

A dual error of $\tilde{\varepsilon}$ shifts each provider's threshold by at most $\tilde{\varepsilon}$, widening the effective noise band from $\bar{\varepsilon} = O(\varepsilon)$ to $\bar{\varepsilon} + \tilde{\varepsilon}$. Steps 3–5 hold with $\bar{\varepsilon} + \tilde{\varepsilon}$ in place of $\bar{\varepsilon}$, contributing an additional welfare gap of $O\Big(\tilde{\varepsilon} \sum_j \bar{c}_j\Big)$. Substituting Lemma 2 directly:

$$O\left(\tilde{\varepsilon} \sum_j \bar{c}_j\right) \;=\; O\left(\varepsilon^2 \sum_j \bar{c}_j\right) + O\left(\sum_j \bar{c}_j \cdot \sqrt{\frac{\log(M/\nu)}{S}}\right).$$

Here, by hypothesis we have that $1/\sqrt{N} \le \varepsilon$, which holds if $N \ge 1/\varepsilon^2$. The first term satisfies $O(\varepsilon^2 \sum_j \bar{c}_j) \le O(\varepsilon \sum_j \bar{c}_j)$ for $\varepsilon \le 1$ and is absorbed into (3).

Step 7. Combining the bounds. From Steps 2–6:

$$\mathrm{OPT} - R(\mathbf{X}^{\mathrm{SAM}}) \;\le\; \big[\mathrm{OPT} - R(\mathbf{X}^{\hat{\Delta},\hat{\lambda}})\big] + \varepsilon_k + O\left(\sum_j \bar{c}_j \sqrt{\frac{\log(M/\nu)}{S}}\right).$$

Substituting and simplifying gives:

$$\mathrm{OPT} - R(\mathbf{X}^{\mathrm{SAM}}) \;\le\; O\left((\varepsilon + \sqrt{\frac{\log(M/\nu)}{S}}) \sum_j \bar{c}_j\right).$$

with probability at least $1-\nu$.